



Flipsky ESC Tool V1.4
User Manual

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1 Overview

Welcome to Flipsky ESC Tool !

This ESC Tool is developed specially for Flipsky FT series ESCs. To get started, you can use it to control the motor, do input settings, firmware upgrade, restore factory default setup and data analysis etc. respectively.

Please read the user guide carefully. Flipsky shall not be liable for any defects that are caused by neglect, misuse or mistreatment by the customer, including improper installation or testing, or for any products that have been altered or modified in any way by the customer. Moreover, Flipsky shall not be liable for any defects that result from the customers design, specifications or instructions for such products. Flipsky deserves the right to change products design, appearance, specifications etc.

2 Operation Requirements

2.1 Hardware Requirement

- FT Series ESCs like FT85BS, FT85BS V2.0, FT85BD, FT85RS, FT60BD, FT85KS. More escs will be launched in the future.
- Type-C USB cable

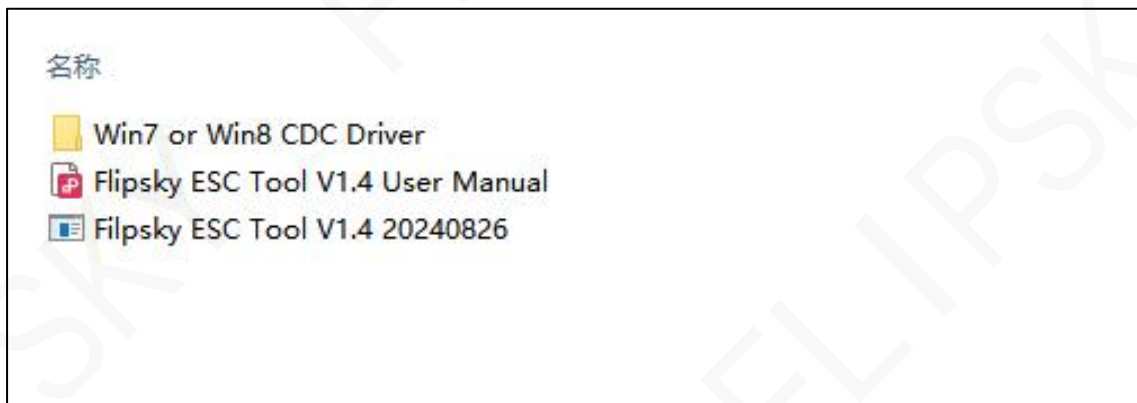
2.2 Computer System Requirement

- Windows 7 or above operating system is required.
- Install a CDC port driver when use Windows 7 or Windows 8 for ESC and ESC Tool recognition and connect.
- No need to install the driver for Windows 10 or above as the system built with it already.

2.2.1 System Driver Download and Related Files Description

Folder Name: Flipsky ESC Tool V1.4 .
Related files as below Screenshot 2-1.

Screenshot 2-1. Flipsky ESC Tool **V1.4**



Win7_or_ Win8 CDC Driver: Windows 7 / Windows 8 CDC port driver installation file

Flipsky ESC Tool V1.4.exe

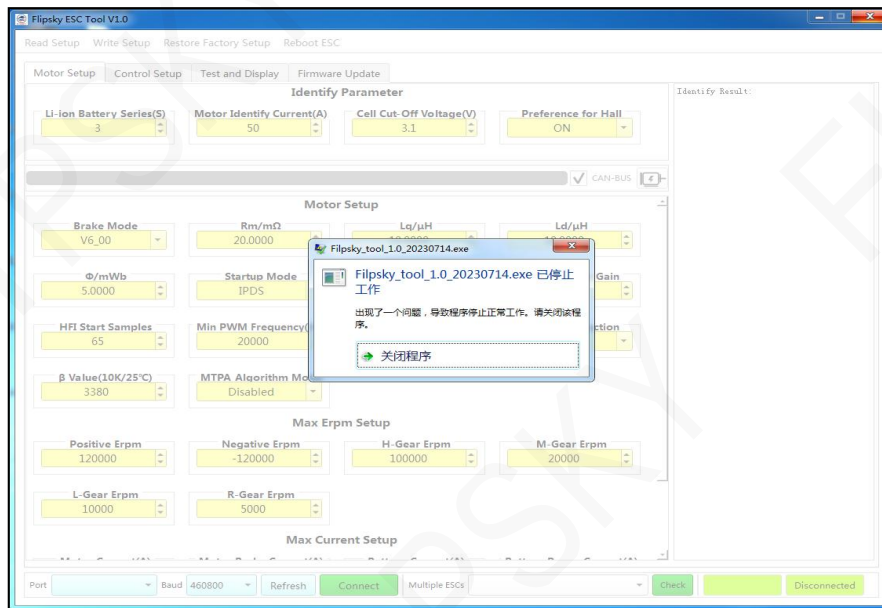
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2.2.2 Driver Update / Reinstall for Windows 7 / Windows 8

Setting Windows 7 as an example:

The ESC cannot connect with the ESC Tool at the first time for no compatible system driver and show as below Screenshot 2-2. Please update / reinstall the port driver for your computer system to connect.

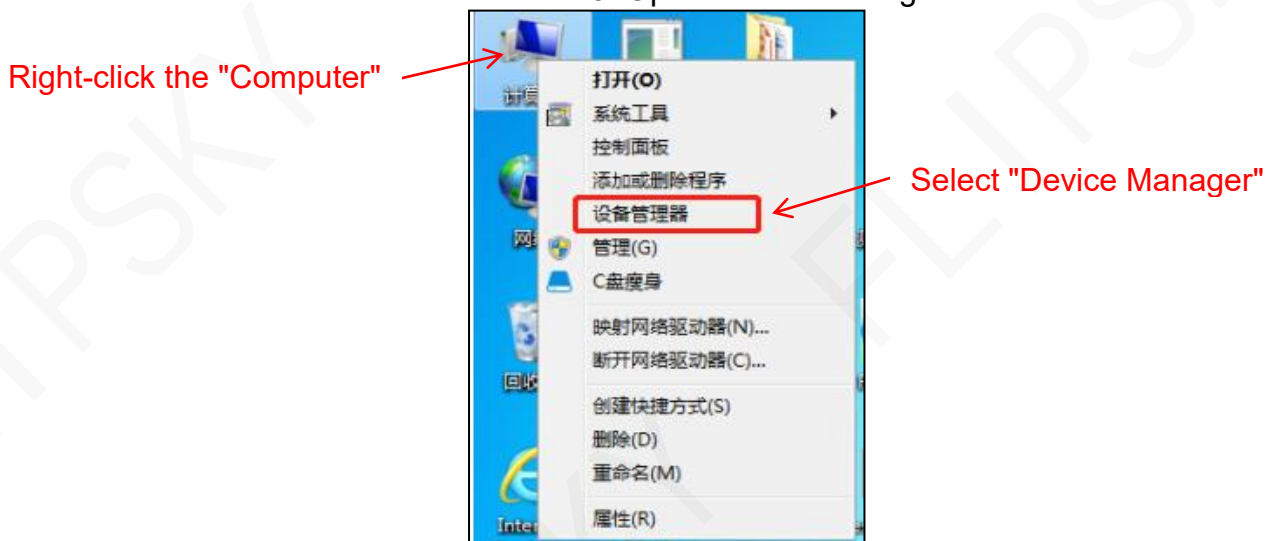
Screenshot 2-2. Fail to connect



Driver Update:

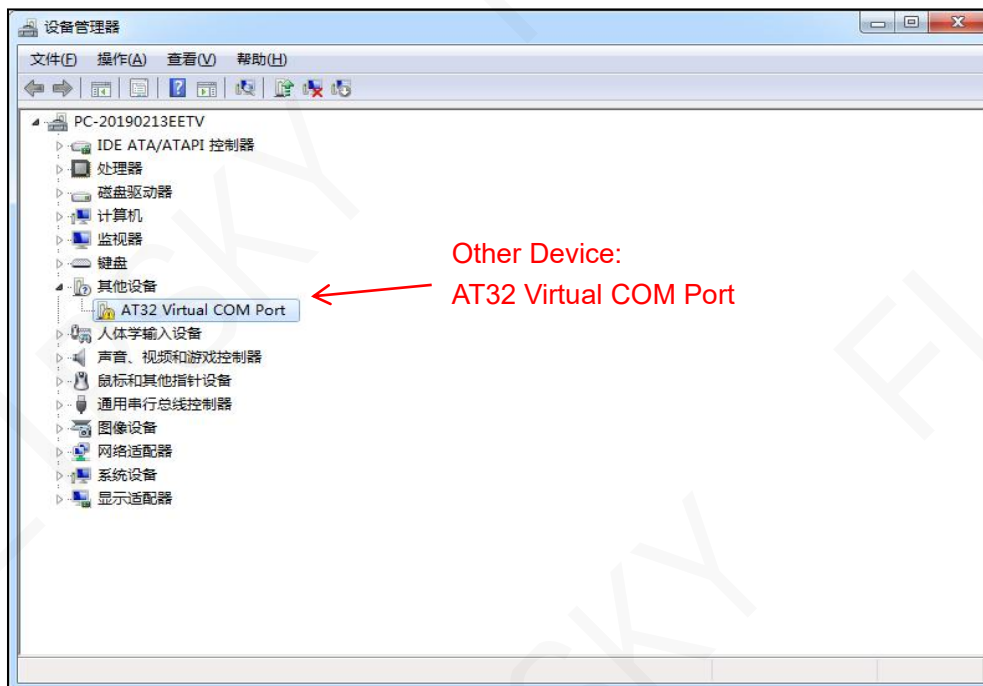
- Connect the ESC with the computer via the USB cable, right-click the "Computer" and select "Device Manager" as below Screenshot 2-3.

Screenshot 2-3. Open Device Manager



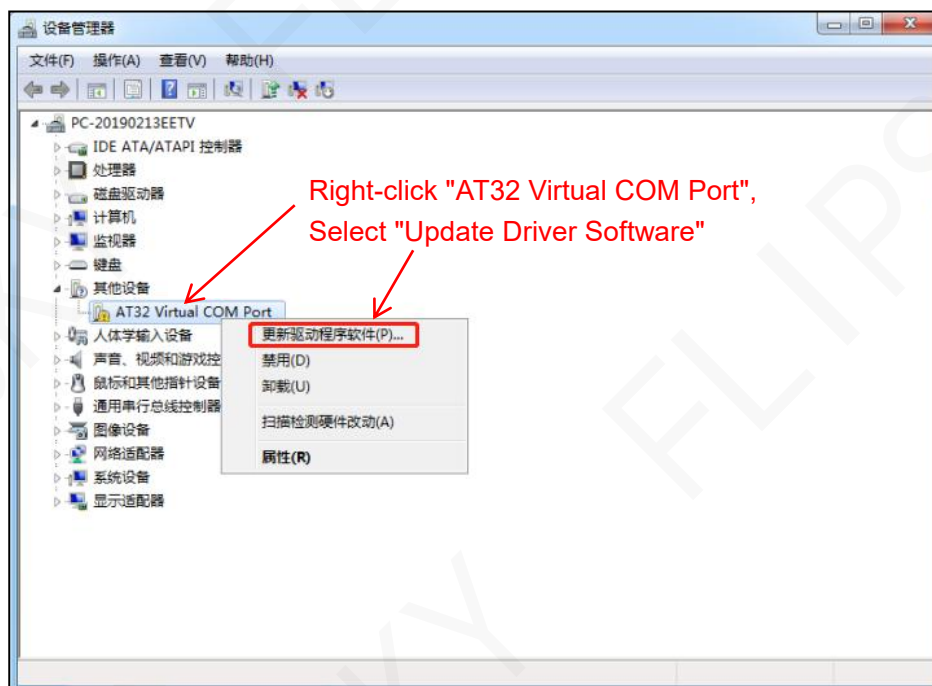
- When the "AT32 Virtual COM Port" with a yellow icon as below Screenshot 2-4, it means that you don't have the latest driver and it needs to be updated.

Screenshot 2-4. COM port driver cannot be used with a yellow icon



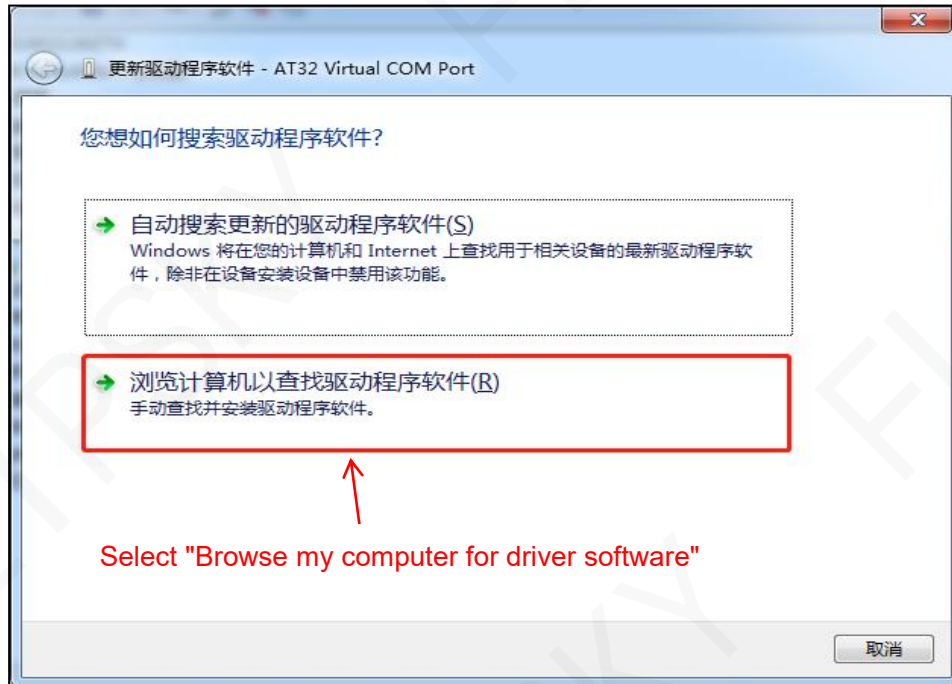
- Right-click the "AT32 Virtual COM Port" and select "Update Driver Software" as below Screenshot 2-5.

Screenshot 2-5. Select "Update Driver Software"



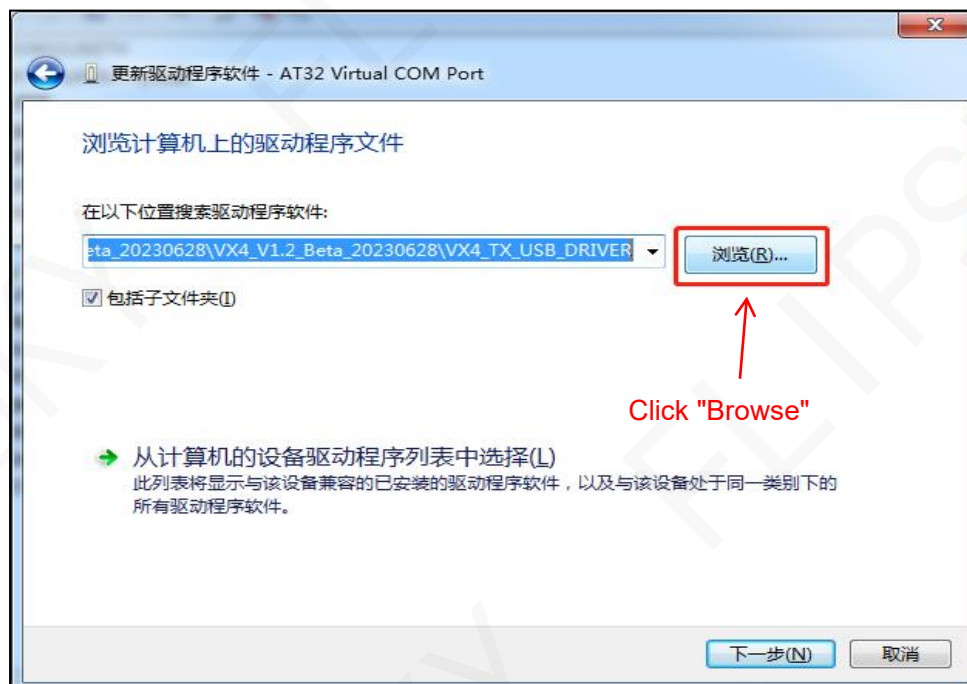
- Select "Browse my computer for driver software" as below Screenshot 2-6.

Screenshot 2-6. Select "Browse my computer for driver software" as below Screenshot 2-6



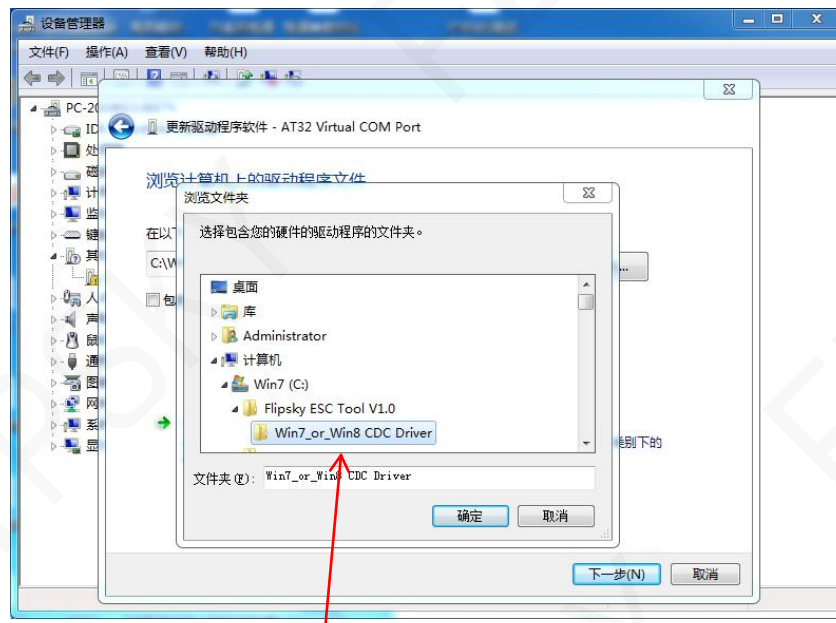
- Click "Browse" to find the driver file as below Screenshot 2-7.

Screenshot 2-7. Click "Browse" to find the location driver file



- Select the location driver installation folder as below example Screenshot 2-8.

Screenshot 2-8. Select the driver installation folder



Find and Select "Win7_or_ Win8 CDC Driver" Folder, and Click "Next" .

- If a security prompt appears, select "Always install this driver software" as below Screenshot 2-9.

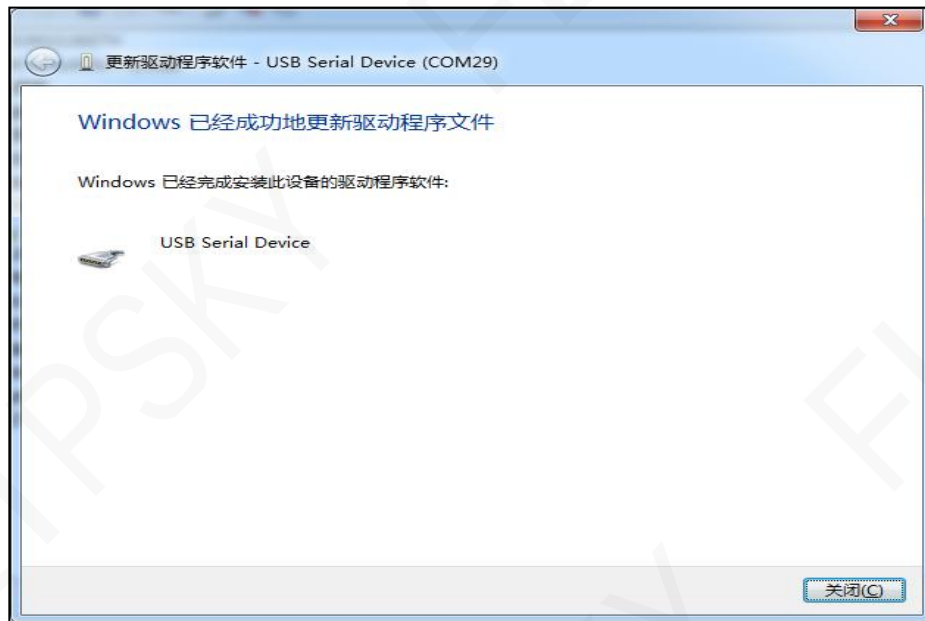
Screenshot 2-9. Select "Always install this driver software"



Select "Always install this driver software"

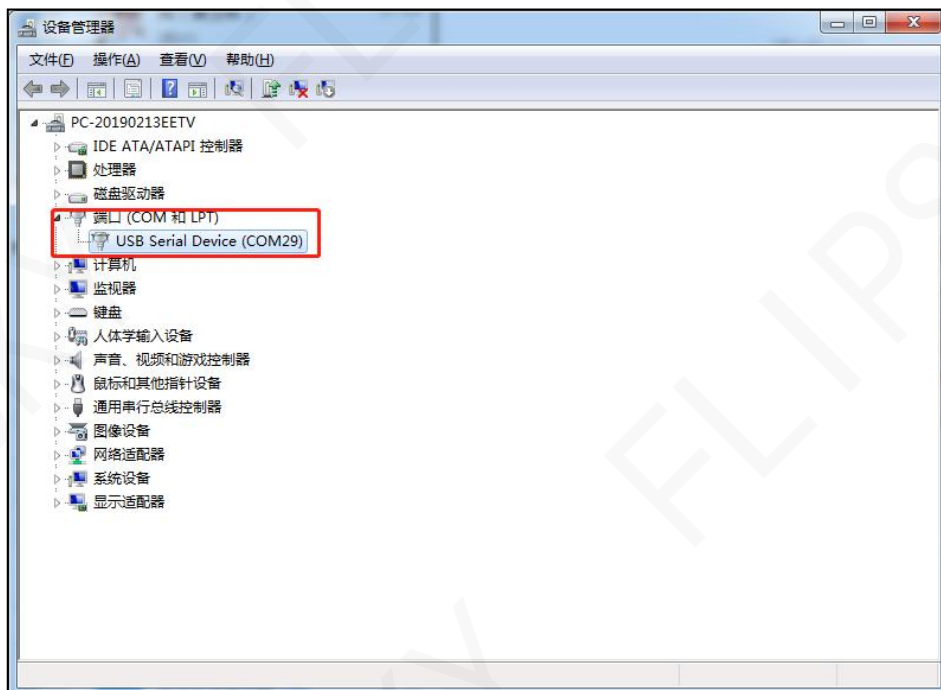
- And then, the driver update finish as below Screenshot 2-10.

Screenshot 2-10. The Driver Installation Finish



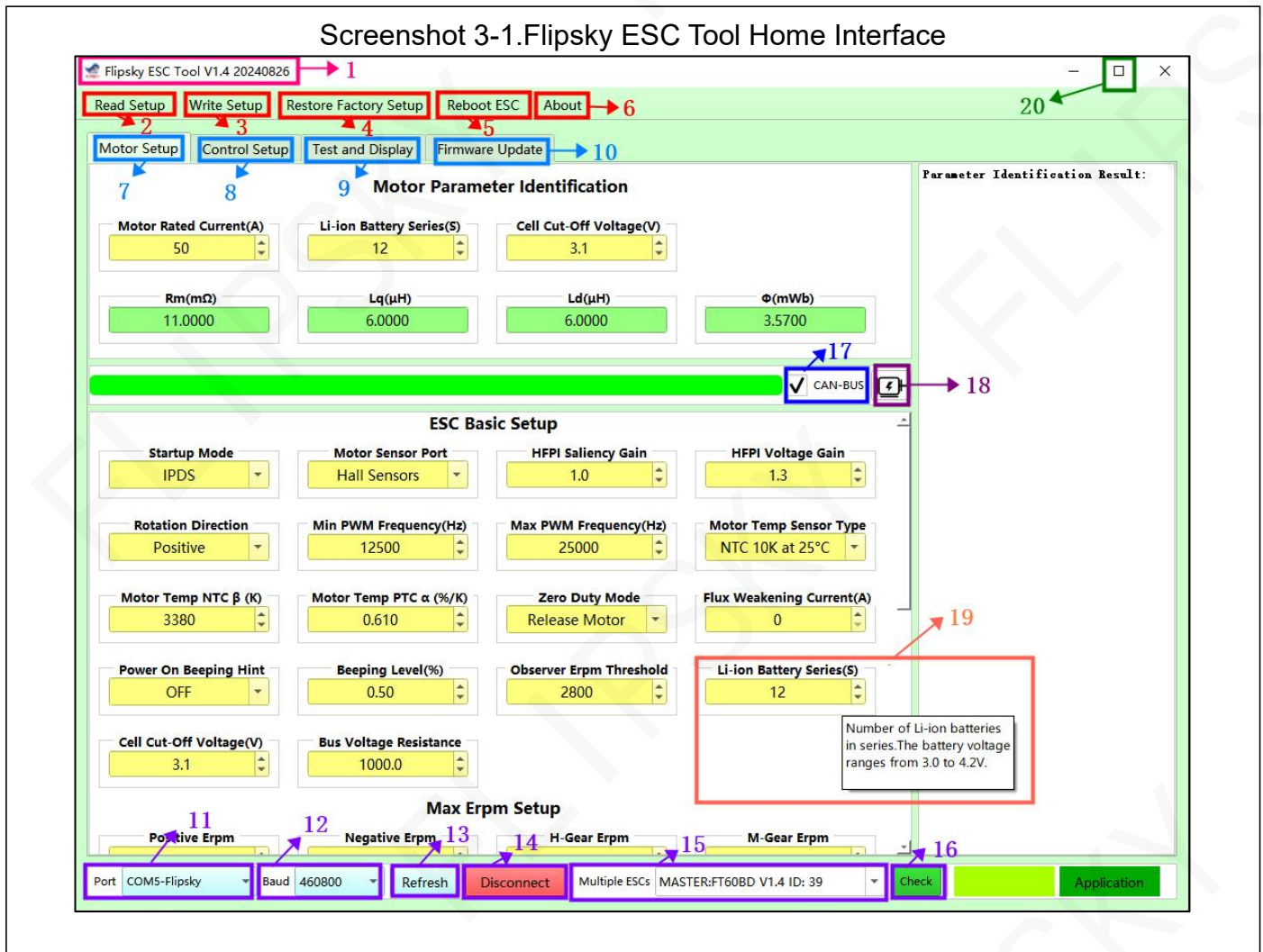
- Enter the "Device Manager" and check if no yellow icon before the USB Serial Device. If so, it means that the driver has been successfully updated as below Screenshot 2-11.

Screenshot 2-11. The USB Serial Device No Yellow Icon



3 Flipsky ESC Tool Home Interface Introduction

Flipsky ESC Tool home interface as below Screenshot 3-1.



- (1) Flipsky ESC Tool V1.2
- (2) Read Setup: Read the current parameters of the ESC
- (3) Write Setup: Write and save the current page parameters
- (4) Restore Factory Setup: Restore factory default setup
- (5) Reboot ESC: Click to re-connect the ESC
- (6) About: Click to read firmware version
- (7) Motor Setup: Click to start the motor FOC settings
- (8) Control Setup: Click to start input control type setup
- (9) Test and Display: Click to start the control test and data analysis
- (10) Firmware Update: Click to start firmware bootloader and update

- (11) Port: COM select
- (12) Baud: Baud rate
- (13) Refresh: Detection refresh port
- (14) Connect / Disconnect: Connect or disconnect the ESC with the ESC Tool
- (15) Multiple ESCs: CAN devices ID check
- (16) Check: Scan all CAN devices
- (17) CAN-BUS: Click to identify multiple motor devices
- (18) Parameters Definitions: Explanation will be displayed when the computer mouse docked the parameters.
- (19) Parameter identification result: click to start the parameters settings.
- (20) Maximize the window available.

4 Connection

4.1 FTESC and Flipsky ESC Tool Connection

Power on the ESC and connect it with the computer via USB. Open the Flipsky ESC Tool V1.4 and the port for ESC ID (COM6-Flipsky) will be displayed. If no ID displays, please click "Refresh" to scan again as below Screenshot 4-1.

Screenshot 4-1. Port ESC ID Display and Refresh



Click "Connect" for ESC and ESC Tool connection. The ESC ID appears as below Screenshot 4-2.

Screenshot 4-2. ESC ID Display



The USB connection ESC is the master board, and all others are slave boards. Theoretically, 1 master board can be connected with a maximum of 253 slave CAN devices (except for the ID number for the master board, 0 and 255, others are fine). While, it is recommended not using more than 8 slave boards in practice since too many CAN devices affect the working performance efficiency. Moreover, please remove the CAN terminal resistors except for the first and last ESCs' CANs. Set 1pcs Master ESC and 6pcs Slave ESCs as an example:

Master(Start)==Slave1==Slave2==Slave3==Slave4==Slave5==Slave6(End)

Remove the CAN terminal resistors of Slave1, Slave2, Slave3, Slave4 and Slave5, leaving only the CAN terminal resistors of Master (Start) and Slave6 (End).

4.2 CAN Devices Connection Problem

If multiple ESCs are connected via CAN but the ESC Tool fails to scan all devices, 3 possible reasons for failure as follows:

- 1) The slave ESC boards' CAN terminal resistors haven't been removed.
- 2) Different ESCs with same IDs
- 3) One of ESCs' CAN baud rate is inconsistent with others.

Troubleshooting:

- 1) Make sure that the ESCs' CAN terminal resistors have been removed correctly except for the Start and the End CAN ESCs (section 4.1).

- 2) Unplug the CAN-BUS wires. Check the ESCs IDs one by one by connecting them with the ESC Tool separately via USB cable. If same IDs happen, revise them to make sure the uniqueness of the ESC ID.

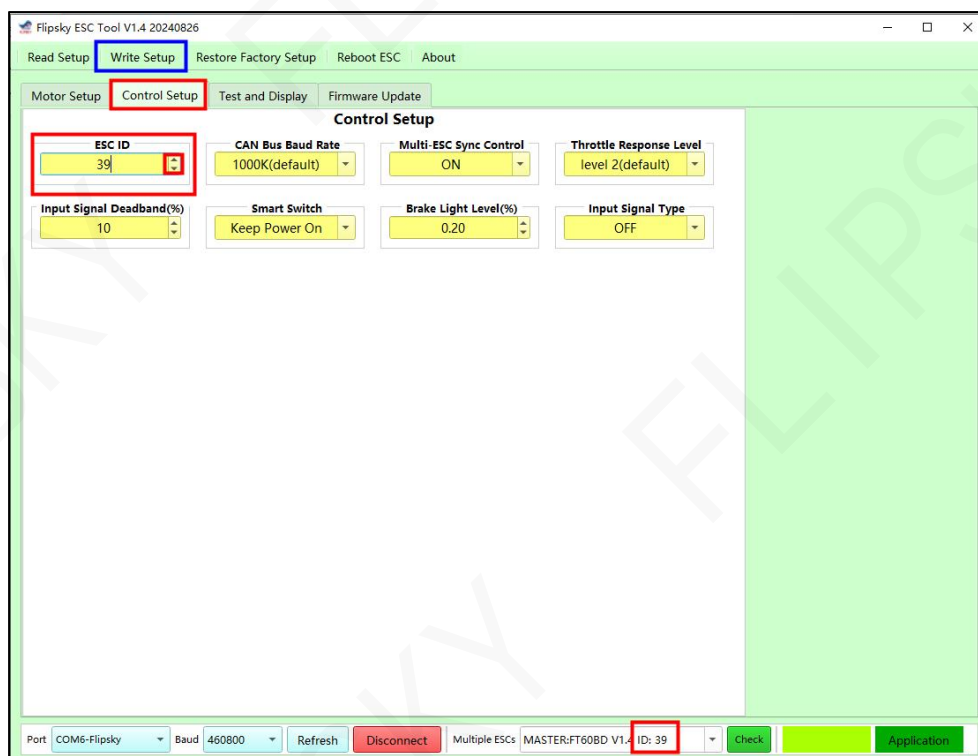
- 3) If still not workable and missing the ESCs, please check the ESC's CAN baud rate and make sure the consistency.

- 4) If still not succeed after the above trials, please restore the factory default setup and try to connect the ESCs again.

4.2.1 How to Set Multiple ESCs ID

The setting steps are shown as below Screenshot 4-3.

Screenshot 4-3.ESC ID Setup



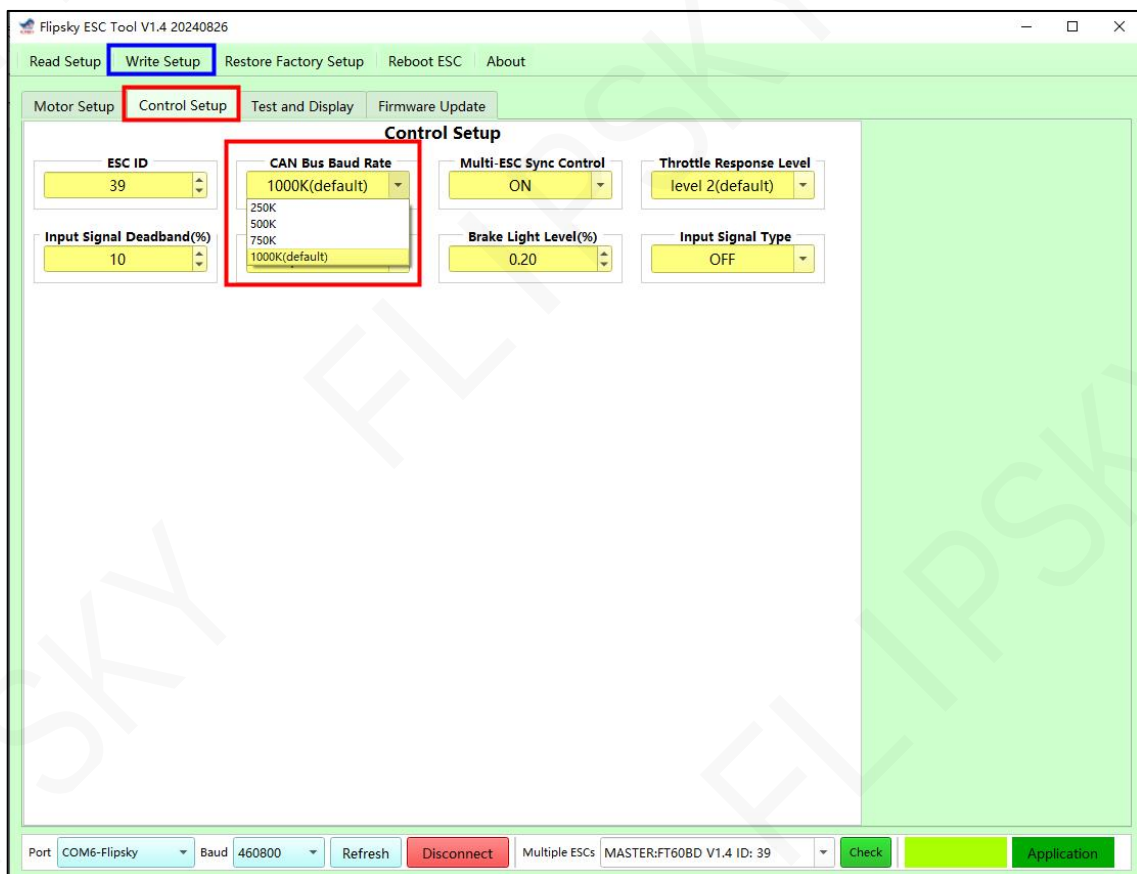
- Click "Control Setup" to enter the control setup interface
- Check the ESC ID and set the ID number from 1 to 254 (see details in section 7.1)
- Click the upper left menu "Write Setup" to save the setup.

4.2.2 How to Set CAN Baud Rate

The setting steps are shown as below Screenshot 4-4.

- Click "Control Setup" to enter the control setup interface
- Check the CAN-Bus Baud Rate (see details in section 7.2)
- Click the upper left menu "Write Setup" to save the setup

Screenshot 4-4. CAN-Bus Baud Rate Setup



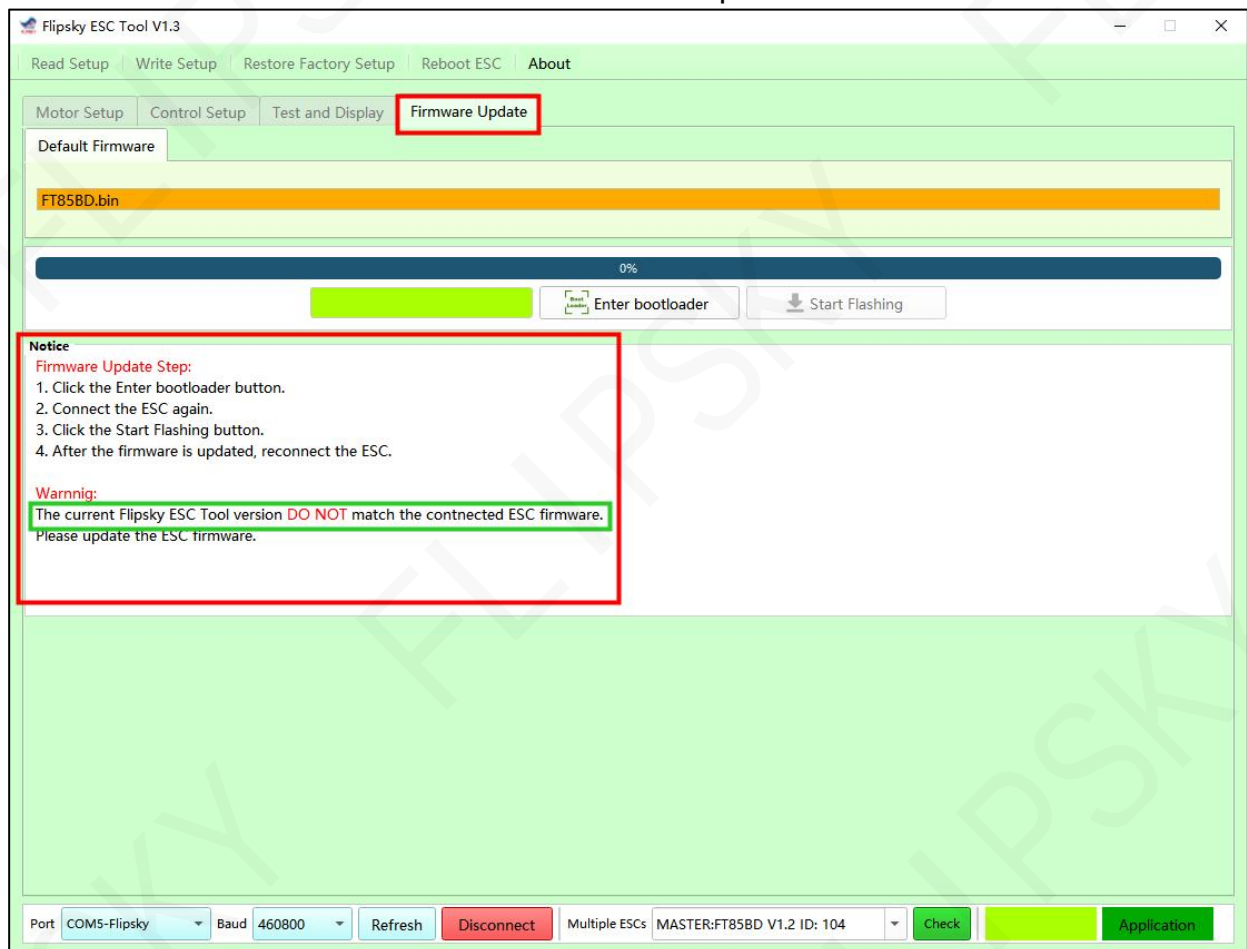
5 Firmware Update

5.1 ESC Firmware Version Check

Click "Firmware Update" to enter the firmware update interface, and it will check if the ESC firmware version matches the Flipsky ESC Tool version automatically. If not matching, please update the firmware as below Screenshot 5-1.

Notes: Please do firmware update for both sides of the dual esc controller via USB cable separately.

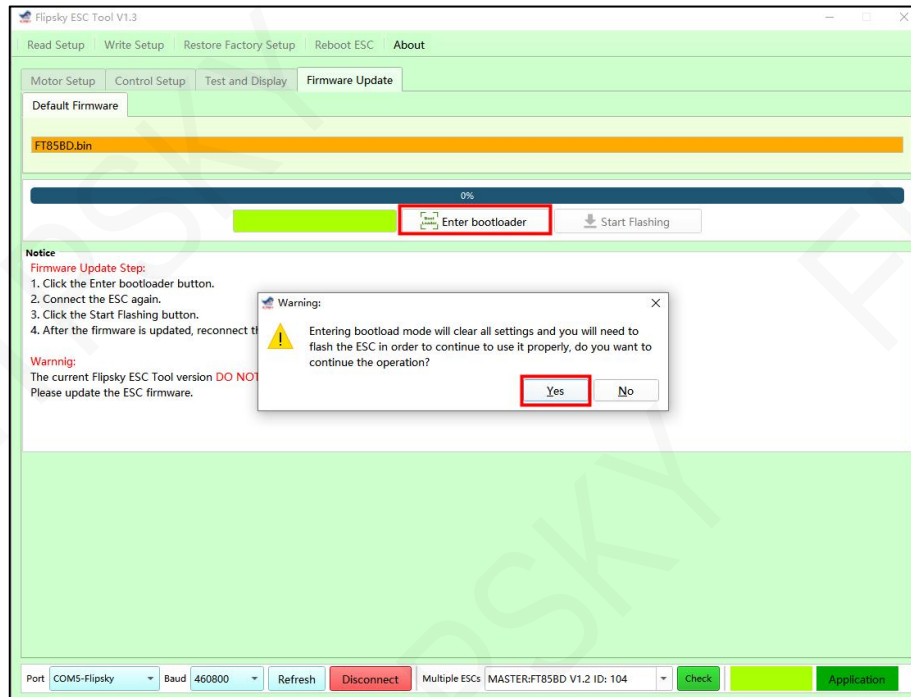
Screenshot 5-1 Firmware Update Interface



5.2 Firmware Update

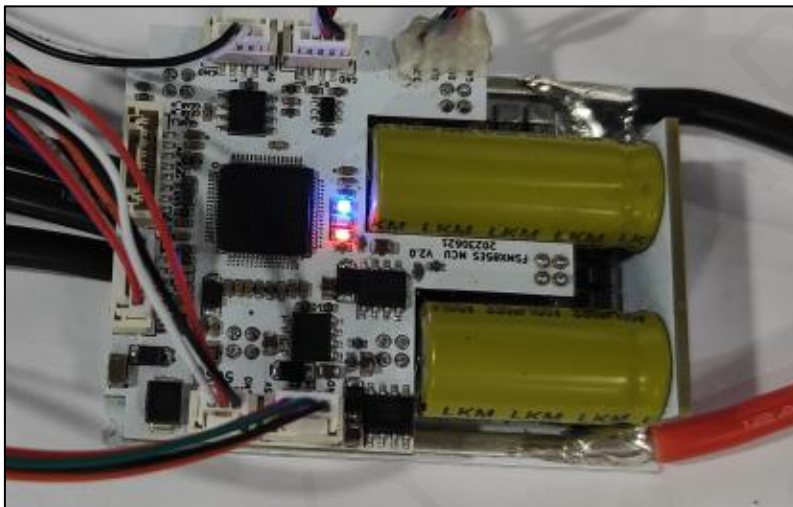
- Step 1: Click "Enter bootloader" and "Yes" to enter the Bootloader mode as below Screenshot 5-2.

Screenshot 5-2. Enter Bootloader Interface

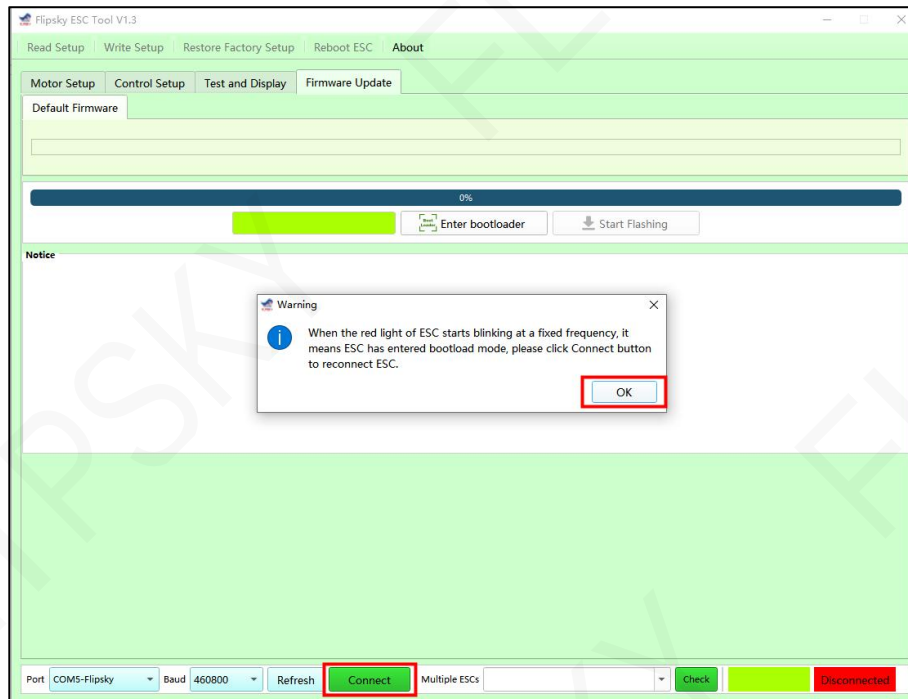


- Step 2: When the ESC red LED blinking at a fixed frequency, it means entering the Bootloader mode successfully (as shown in Screenshot 5-3). Click "OK" and "Connect" to reconnect the ESC as below Screenshot 5-4.

Screenshot 5-3. ESC entering the bootloader mode

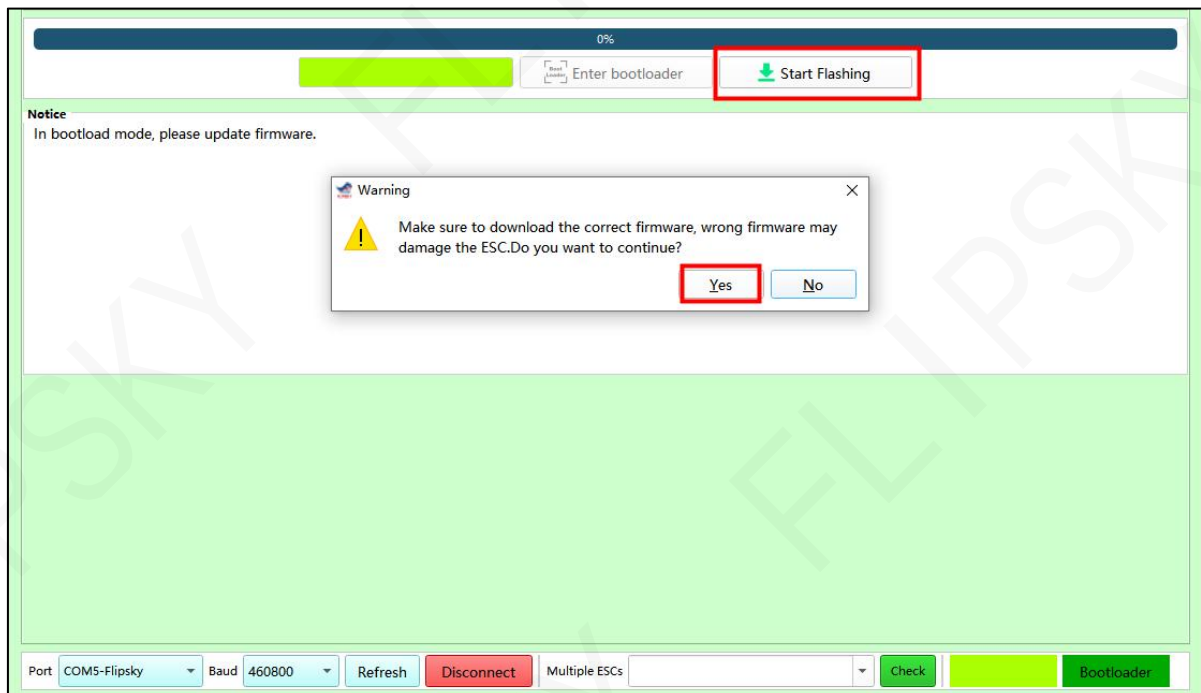


Screenshot 5-4. Reconnect the ESC

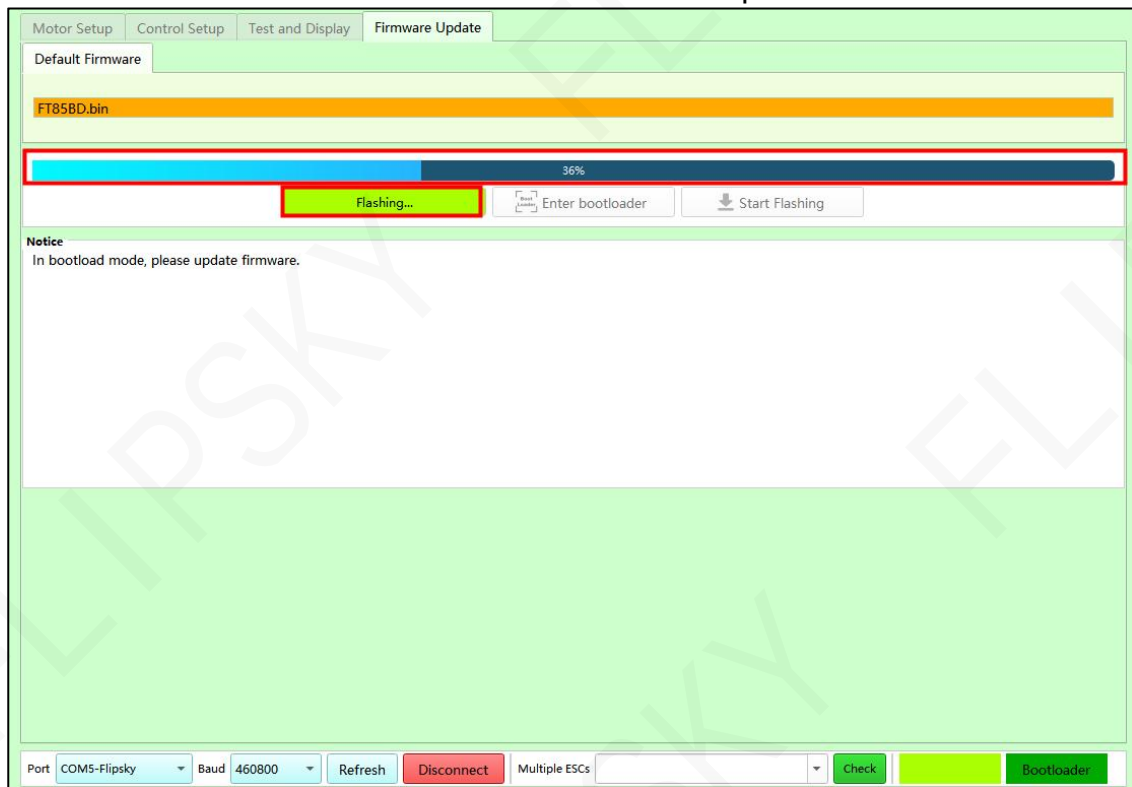


- Step 3: Click "Start Flashing" and "Yes" to update the firmware as below Screenshot 5-5 and Screenshot 5-6.

Screenshot 5-5. Start Flashing to update firmware

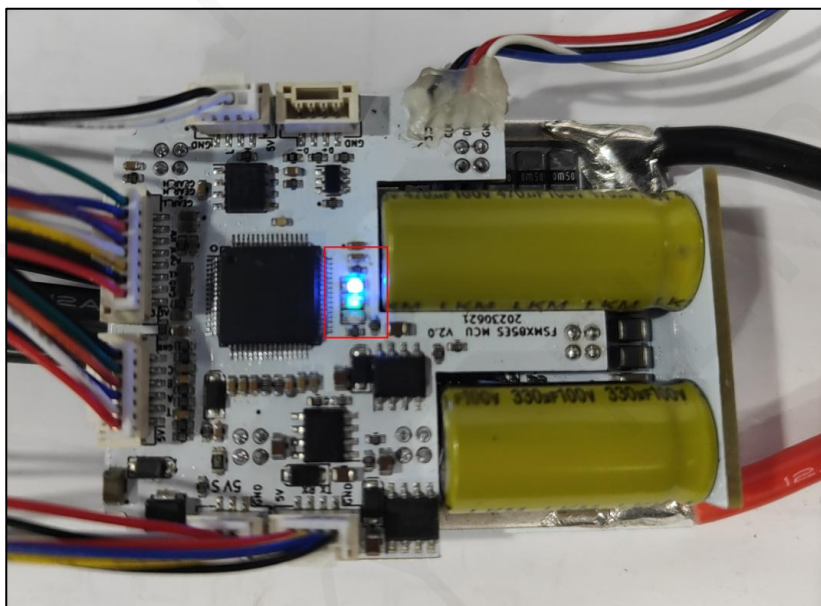


Screenshot 5-6. Firmware Update



- When the ESC's green and blue LEDs are solid on after the blinking of red and green LEDs, it means the finish of firmware update (as Screenshot 5-7). Click "Connect" to reconnect the ESC.

Screenshot 5-7. Solid blue and green led on



6 Motor Setup

Motor setup interface is in home interface as Screenshot 6-1.

Screenshot 6-1. Motor Setup Interface

The screenshot displays the 'Motor Setup' interface of the Flipsky ESC Tool V1.4. The window title is 'Flipsky ESC Tool V1.4 20240826'. The top menu bar includes 'Read Setup', 'Write Setup', 'Restore Factory Setup', 'Reboot ESC', and 'About'. The 'Motor Setup' tab is selected and highlighted with a red box. Below the tabs, the 'Motor Parameter Identification' section contains several input fields: 'Motor Rated Current(A)' (50), 'Li-ion Battery Series(S)' (6), 'Cell Cut-Off Voltage(V)' (3.1), 'Rm(mΩ)' (17.5327), 'Lq(μH)' (21.3481), 'Ld(μH)' (15.0179), and 'Φ(mWb)' (4.1849). A green bar with a checkmark and 'CAN-BUS' icon is visible. The 'ESC Basic Setup' section includes: 'Startup Mode' (IPDS), 'Motor Sensor Port' (Hall Sensors), 'HFPI Saliency Gain' (1.0), 'HFPI Voltage Gain' (1.3), 'Rotation Direction' (Positive), 'Min PWM Frequency(Hz)' (8010), 'Max PWM Frequency(Hz)' (10562), 'Motor Temp Sensor Type' (NTC 10K at 25°C), 'Motor Temp NTC β (K)' (3380), 'Motor Temp PTC α (%/K)' (0.610), 'Zero Duty Mode' (Release Motor), 'Flux Weakening Current(A)' (0), 'Power On Beeping Hint' (OFF), 'Beeping Level(%)' (0.50), 'Observer Erpm Threshold' (3704), 'Li-ion Battery Series(S)' (6), 'Cell Cut-Off Voltage(V)' (3.1), and 'Bus Voltage Resistance' (1000.0). The 'Max Erpm Setup' section at the bottom has four tabs: 'Positive Erpm', 'Negative Erpm', 'H-Gear Erpm', and 'M-Gear Erpm'. The bottom status bar shows 'Port: COM6-Flipsky', 'Baud: 460800', 'Refresh', 'Disconnect', 'Multiple ESCs', 'MASTER:FT60BD V1.4 ID: 39', 'Check', and 'Application' buttons.

6.1 Parameters Identification Wizard

6.1.1 Preparations for Parameters Identification

Set "Motor Rated Current(A)", "Li-ion Battery Series(S)" and "Cell Cut-Off Voltage" before doing parameters identification as below Screenshot 6-2.

Screenshot 6-2. Three parameter settings before identification

The screenshot displays the Flipsky ESC Tool V1.4 interface. The 'Motor Setup' tab is active, and the 'Motor Parameter Identification' section is highlighted. The settings shown are:

Parameter	Value
Motor Rated Current(A)	50
Li-ion Battery Series(S)	6
Cell Cut-Off Voltage(V)	3.1
Rm(mΩ)	17.5327
Lq(μH)	21.3481
Ld(μH)	15.0179
Φ(mWb)	4.1849

Below the Motor Parameter Identification section is the 'ESC Basic Setup' section, which includes settings for Startup Mode (IPDS), Motor Sensor Port (Hall Sensors), HFPI Saliency Gain (1.0), HFPI Voltage Gain (1.3), Rotation Direction (Positive), Min PWM Frequency (8010 Hz), Max PWM Frequency (10562 Hz), Motor Temp Sensor Type (NTC 10K at 25°C), Motor Temp NTC β (K) (3380), Motor Temp PTC α (%/K) (0.610), Zero Duty Mode (Release Motor), Flux Weakening Current (A) (0), Power On Beeping Hint (OFF), Beeping Level (%) (0.50), Observer Erpm Threshold (3704), and Li-ion Battery Series (S) (6). At the bottom, there's a 'Max Erpm Setup' section with Positive Erpm, Negative Erpm, H-Gear Erpm, and M-Gear Erpm. The status bar at the bottom shows Port: COM6-Flipsky, Baud: 460800, and buttons for Refresh, Disconnect, Check, and Application.

- Motor Rated Current(A): Write the motor rated current per motor specifications, not the max. or peak current (default: 50A; adjustable range: 5-500A)
- Li-ion Battery Series(S): Write the battery Lipos per battery specifications (1S=4.2V full charge; default: 3S; adjustable range: 3-30S)
- Cell Cut-Off Voltage(V): Battery voltage cutoff per cell setup to prevent the battery over discharge (default: 3.1V; adjustable range: 2.6 ~ 3.3V).

6.1.2 Parameters Identification Steps

- Step 1: Set "Motor Rated Current(A)", "Li-ion Battery Series(S)", and "Cell Cut-Off Voltage(V)" accordingly (See Screenshot 6-2).
- Step 2: Remember to tick "CAN-BUS" when doing settings for dual or multiple ESCs and drivers as below Screenshot 6-3.

Screenshot 6-3 CAN-BUS

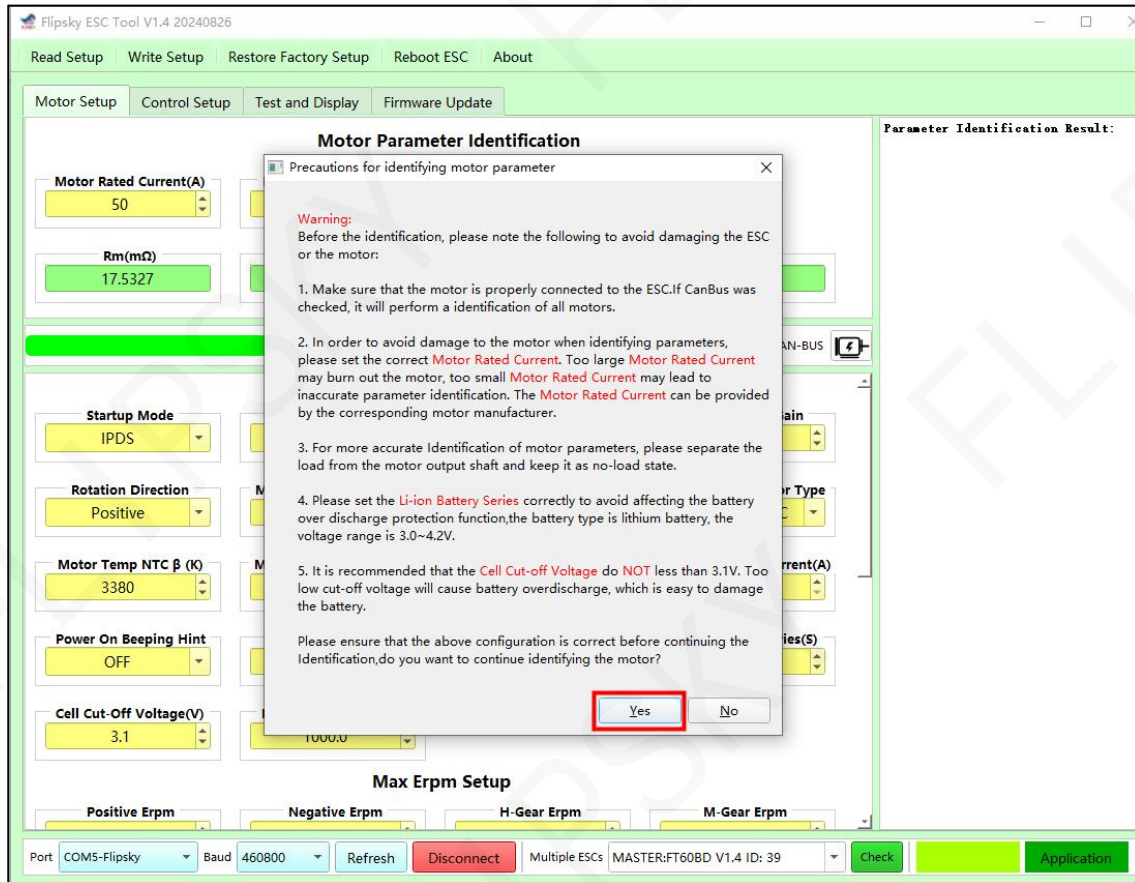
The screenshot shows the 'Motor Parameter Identification' window. It contains several input fields for motor parameters: Motor Rated Current(A) set to 50, Li-ion Battery Series(S) set to 3, Cell Cut-Off Voltage(V) set to 3.1, Rm(mΩ) set to 11.0000, Lq(μH) set to 6.0000, Ld(μH) set to 6.0000, and Φ(mWb) set to 3.5700. At the bottom right, there is a checkbox labeled 'CAN-BUS' which is checked, and a button with a battery icon.

- Step 3: Click the Parameter Identification Button (as shown in Screenshot 6-4) and the notification window popup "Precautions for identifying motor parameter". Click "Yes" to proceed next step as below Screenshot 6-5.

Screenshot 6-4 Parameter Identification Button

This screenshot is identical to Screenshot 6-3, showing the 'Motor Parameter Identification' window with the same parameter values. However, the button with the battery icon at the bottom right is highlighted with a red box, indicating it is the 'Parameter Identification Button' mentioned in the text.

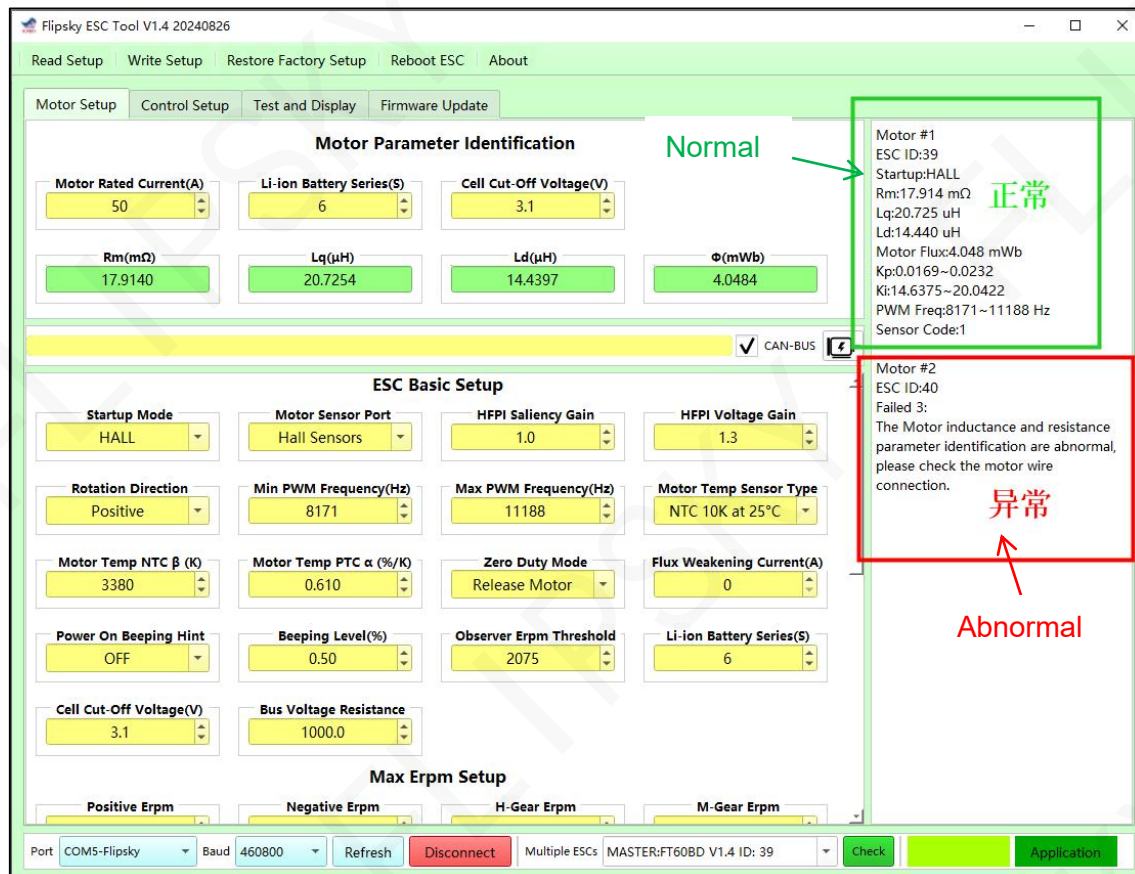
Screenshot 6-5 Precautions for identifying motor parameter



6.1.3 Motor Parameter Identification Results and Fault Codes

Please check the results on the right side of the home page (as shown in Screenshot 6-6) after the finish of motor Parameter Identification to see if all data are normal or not. If abnormal, please take corrective actions per fault codes and failed reasons notification.

Screenshot 6-6. Parameter identification results



Normal interpretation of motor parameter identification results:

- ESC ID: ESC ID Number
- Startup: Motor Starting Mode
 - ✧ IPDS: Initial Position Detection Startup (sensorless mode)
 - ✧ HFPI: High Frequency Pulse Injection (sensorless mode)
 - ✧ HALL: Hall Sensor Mode
- Rm: Motor Phase Resistance (mΩ)
- Lq: Motor Q-Axis Inductance (uH)
- Ld: Motor D-Axis Inductance (uH)
- Motor Flux(Φ): Motor Flux Linkage (mWb)
- Kp、Ki: Dynamic variation range of Proportional(Kp) and Integral(Ki)

- PWM Freq: Dynamic variation range of PWM Frequency
- Hall Code:
 - ✧ Hall Code: 1→HALL Sensor
 - ✧ Hall Code: 2→One Hall Phase Absence
 - ✧ Hall Code: 3→Two Hall Phase Absence
 - ✧ Hall Code: 4→Hall Three-Phase Short Circuit
 - ✧ Hall Code: 5→Overloading, Undetected
 - ✧ Hall Code: 6→No HALL Sensor Connection

Interpretation of motor parameters identification abnormal results:

- Failed 1: The battery Lipos number is set incorrectly, and does not match the actual operating voltage. Please set the correct number of battery Lipos.
Notes: Please make sure the battery Lipos matching the using battery. Otherwise, it leads to detection failure, battery over-discharge protection, battery life span etc.
- Failed 2: The ESC is still faulty, please check the error code.
Notes: Please check the fault code and take corrective actions per codes notification.
- Failed 3: The Motor inductance and resistance parameter identification are abnormal, please check the motor wire connection.
Notes: Please check the ESC and motor wires connections.
- Failed 4: The motor hall sensor identify failed. please check the motor hall sensor.
Notes: Please check the motor hall sensor wire connection and measure the motor hall sensor voltage to see if normal.
- Failed 5: The motor flux linkage identify failed. Maybe the motor load is too heavy.
Notes: Please check if overloading happens or the correct matching motor setup.

6.2 ESC Parameters Definitions and Configuration

Some motor parameters like R_m , L_q , L_d , Φ , Startup Mode、HFPI Saliency Gain、HFPI Voltage Gain、Max PWM Frequency、Min PWM Frequency etc. will be calculated automatically after the parameter identification finish which don't need to change manually. As for Max Erpm and Max Current, please write per application requirements as below Screenshot 6-7.

Notes: Remember to click the upper left button "Write Setup" to save settings.

Screenshot 6-7. Adjustable parameters

The screenshot displays the Flipsky ESC Tool V1.4 interface. The 'Motor Setup' tab is active, and the 'ESC Basic Setup' section is highlighted with a red box. The interface includes the following sections:

- Motor Parameter Identification:**
 - Motor Rated Current(A): 50
 - Li-ion Battery Series(S): 6
 - Cell Cut-Off Voltage(V): 3.1
 - $R_m(m\Omega)$: 17.7850
 - $L_q(\mu H)$: 20.4922
 - $L_d(\mu H)$: 13.0699
 - $\Phi(mWb)$: 4.0697
- ESC Basic Setup:**
 - Startup Mode: IPDS
 - Motor Sensor Port: Hall Sensors
 - HFPI Saliency Gain: 1.0
 - HFPI Voltage Gain: 1.3
 - Rotation Direction: Positive
 - Min PWM Frequency(Hz): 8178
 - Max PWM Frequency(Hz): 11916
 - Motor Temp Sensor Type: NTC 10K at 25°C
 - Motor Temp NTC β (K): 3380
 - Motor Temp PTC α (%/K): 0.610
 - Zero Duty Mode: Release Motor
 - Flux Weakening Current(A): 0
 - Power On Beeping Hint: OFF
 - Beeping Level(%): 0.50
 - Observer Erpm Threshold: 2114
 - Li-ion Battery Series(S): 6
 - Cell Cut-Off Voltage(V): 3.1
 - Bus Voltage Resistance: 1000.0
- Max Erpm Setup:**
 - Positive Erpm: [Input field]
 - Negative Erpm: [Input field]
 - H-Gear Erpm: [Input field]
 - M-Gear Erpm: [Input field]

On the right side, there is a status panel for Motor #1 and Motor #2. Motor #1 shows successful identification parameters. Motor #2 shows a 'Failed 3' status with a message: 'The Motor inductance and resistance parameter identification are abnormal, please check the motor wire connection.'

At the bottom, there are buttons for 'Read Setup', 'Write Setup', 'Restore Factory Setup', 'Reboot ESC', and 'About'. The 'Write Setup' button is highlighted with a blue box. The bottom status bar shows the port (COM21-Flipsky), baud rate (460800), and a 'Check' button.

6.2.1 ESC Basic Parameters

ESC Basic Setup besides Startup Mode, HFPI Saliency Gain, HFPI Voltage Gain Volt, Rotation Direction, Min PWM Frequency, Max PWM Frequency, Motor Temp NTC β (K), Zero Duty Mode, adding more settings such as Motor sensor Port, Motor Temp sensor Type, Motor Temp PTC α (%/K), Flux Weakening Current(A), Power on Beeping Hint, Beeping Level(%), observer Erpm Threshold, Li-ion Battery series(S), Cell cut-off Voltage(V) and Bus Voltage Resistance as screenshot 6-8.

Screenshot 6-8. ESC Basic Parameters

ESC Basic Setup			
Startup Mode IPDS	Motor Sensor Port Hall Sensors	HFPI Saliency Gain 1.0	HFPI Voltage Gain 1.3
Rotation Direction Positive	Min PWM Frequency(Hz) 10762	Max PWM Frequency(Hz) 14841	Motor Temp Sensor Type NTC 10K at 25°C
Motor Temp NTC β (K) 3380	Motor Temp PTC α (%/K) 0.610	Zero Duty Mode Release Motor	Flux Weakening Current(A) 0
Power On Beeping Hint OFF	Beeping Level(%) 0.50	Observer Erpm Threshold 1647	Li-ion Battery Series(S) 6
Cell Cut-Off Voltage(V) 3.1	Bus Voltage Resistance 1000.0		

● Startup Mode

Three motor startup modes can be selected:

- ✧ IPDS: Initial Position Detection Startup (suitable for light loading sensorless mode such as underwater DPVs, multi-axis FPVs etc.)
- ✧ HFPI: High Frequency Pulse Injection (suitable for heavy loading sensorless mode such as E-Bike, E-Board, Go-cart etc.)
- ✧ HALL: Hall Sensored Mode (widely used, especially suitable for equipments with full load)

Note: HFPI mode is not suitable for all motors. If there are serious vibrations, stalls etc. when using this mode, it means that this mode is not suitable for the motor. If the motor integrates HALL interface, we strongly recommend using HALL for smooth running and better performance.

● Motor sensor Port

Before doing settings, please choose the corresponding motor sensor port and remember to click "Write Setup".

- ✧ Hall sensor: It's a three-way Hall effect position sensor which is mounted every 120°.
- ✧ ABI Encoder: An incremental encoder is a linear or rotary electromechanical device that has two output signals, A and B, which issue pulses when the device is moved.

Note: Since the incremental encoder cannot determine the initial position of the rotor, it is recommended to rotate the motor manually one circle for zero position calibration after the first start of the ESC, otherwise unexpected situations may occur when the control command is forcibly sent.

- ✧ AS5047(SPI): An absolute encoder maintains position information when power is removed from the encoder, which will immediately report changes and no need to do zero position calibration.

● HFPI Saliency Gain

Adjustable range: 0.5~4.0

Parameter description: This parameter setup is designed for the accuracy of angle measurement and reduce noise in HFPI mode, which can be matched automatically after the motor parameter identification finish and it's not necessary to write manually.

● HFPI Voltage Gain

Adjustable range: 0.2~3.0

Parameter description: This parameter setup is designed for high-frequency pulse voltage injection needs of motor D-axis in HFPI mode, which is determined by the motor specification and can be matched automatically after the motor parameter identification finish, so it's not necessary to write manually.

● Rotation Direction

- ✧ Positive
- ✧ Negative (Reverse)

● Min PWM Frequency

Adjustable range: 2000~25000

Parameter description: This parameter can be matched automatically after the motor parameter identification finish and no need to write manually.

● Max PWM Frequency

Adjustable range: 10000~25000

Parameter description: This parameter can be matched automatically after the motor parameter identification finish and no need to write manually.

● Motor Temp sensor Types:

- ✧ NTC 10K at 25°C
- ✧ NTC 10K at 25°C
- ✧ NTC 100K at 25°C
- ✧ PT1000PTC 1K at 100°C
- ✧ KTY83-122
- ✧ KTY83-152
- ✧ KTY84-130
- ✧ KTY84-150
- ✧ KTY84-151
- ✧ KTY84-152

Parameter description: NTC 10K temperature sensor with a temperature of 25°C is the most widely used one in applications but some others have been also used in special application fields.

● Motor Temp NTC β (K)

Adjustable range: 1~10000

Parameter description: A thermistor's " β " value, or beta value, is an indication of the shape of the curve representing the relationship between resistance and temperature of an NTC thermistor. Please write the correct data per your motor. Flipsky's motor NTC β is 3380.

● Motor Temp PTC α (%/K)

Adjustable range: 0.5~100

Parameter description: A PTC thermistor is a thermally sensitive resistor for which the resistance increases significantly with increasing temperature, used as motor protection sensors to protect the motors from overheating. It's important to write the correct α data per your motor.

● Zero Duty Mode

- ✧ Keep Output
- ✧ Release Motor

Parameter description: If set duty cycle 0% in Zero Duty Mode, customers can choose the mode of Keep Output or Release Motor.

- **Flux Weakening Current**

Adjustable range: 0~500

Parameter description: It stands for the current value applied to the D axis during Flux weakening control. Increasing this value can increase the maximum speed limit of the motor. The larger the value, the higher the maximum speed, but the current consumption is also greater and the motor becomes hotter at the same time. It is recommended to set it to no more than 1/4 of the motor max current.

- **Power on Beeping Hint**

Parameter description: After enabling this function, 3 times “Beep” after ESC is powered on. It's workable only after finishing motor parameters setup, otherwise the sound will be abnormal.

- **Beeping Level(%)**

Adjustable range: 0~1

Parameter description: The larger of the value, the louder of the sound.

- **Observer Erpm Threshold**

Adjustable range: 1~250000

Parameter description: When ERPM is bigger than this value, it will switch to the flux observer to measure the rotor position and speed.

- **Li-ion Battery series(s)**

Adjustable range: 3~30S

Parameter description: Generally 1S=4.2V (full charge), please set the lipos per your battery specification.

- **Cell cut-off Voltage(V)**

Adjustable range: 2.6~3.3V

Parameter description: Setting cell cut-off voltage to prevent the battery from over-discharging, and then lengthening the battery life span. Default value: 3.1V.

- **Bus Voltage Resistance**

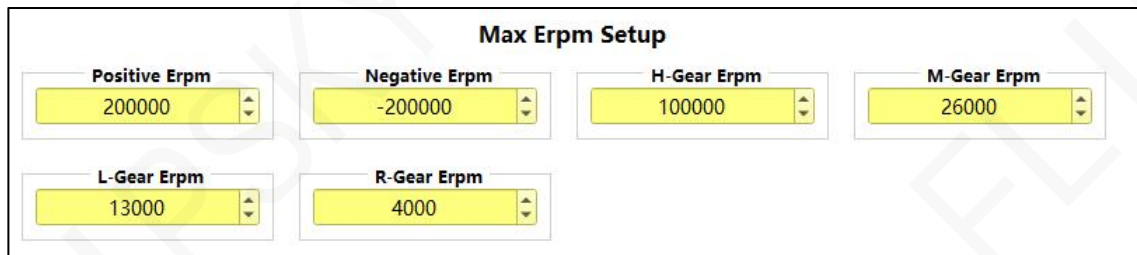
Adjustable range: 0~2000

Parameter description: Adding motor “Bus Voltage Resistance” setup to make the power supply voltage detected more accurately (adjust according to the voltage measured by the multimeter or the power supply voltage). If not necessary, keep the default value without modification.

6.2.2 Max Erpm Setup

Max Erpm Setup includes "Positive Erpm", "Negative Erpm", "H-Gear Erpm", "M-Gear Erpm", "L-Gear Erpm", "R-Gear Max Erpm" as below Screenshot 6-9.

Screenshot 6-9. Max Erpm Setup



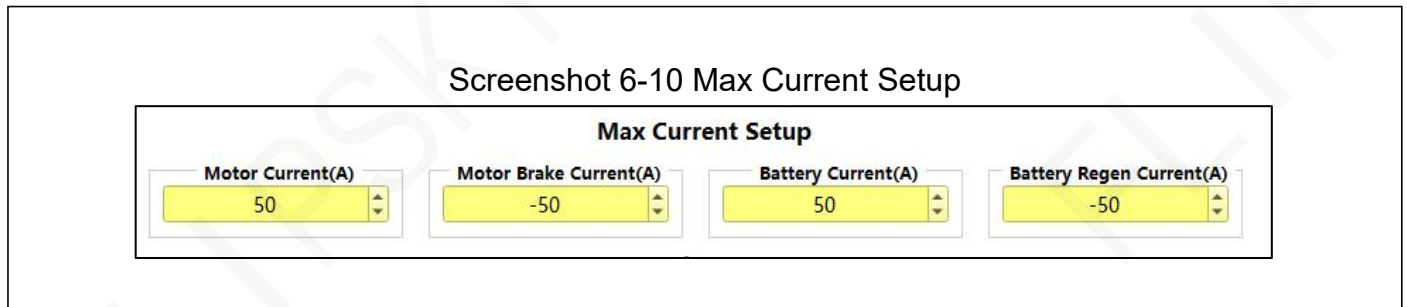
The screenshot displays the 'Max Erpm Setup' window with the following values:

Parameter	Value
Positive Erpm	200000
Negative Erpm	-200000
H-Gear Erpm	100000
M-Gear Erpm	26000
L-Gear Erpm	13000
R-Gear Erpm	4000

- **Positive Max Erpm:** Motor forward maximum Erpm
Adjustable range: 500~250000
- **Negative Max Erpm:** Motor reverse maximum Erpm
Adjustable range: -250000~-500
- **H-Gear Max Erpm:** The motor maximum Erpm at high speed gear
Adjustable range: 500~250000
- **M-Gear Max Erpm:** The motor maximum Erpm at medium speed gear
Adjustable range: 500~250000
- **L-Gear Max Erpm:** The motor maximum Erpm at low speed gear
Adjustable range: 500~250000
- **R-Gear Max Erpm:** The motor maximum Erpm in reverse direction
Adjustable range: 500~250000

6.2.3 Max Current Setup

The max current setup includes both the motor and the battery maximum current setup. The motor maximum current settings include "Motor Current" and "Motor Brake Current". Also, the battery maximum current settings include "Battery Current" and "Battery Regen Current" as below Screenshot 6-10.



Note: The max battery / motor current of each FT series esc are different.

- **Max Motor Current:** The motor maximum continuous current (A)

Adjustable range: 0~500

- **Max Motor Brake Current:** The motor maximum braking current (A)

Adjustable range: -500~0

Notes: If using the DC power supply, the Max Motor Brake Current must be set to "0". Otherwise, the large back EMF will damage the ESC or even the power supply when the motor brakes in a high speed state.

- **Max Battery Current:** Battery maximum continuous output current (A)

Adjustable range: 0~500

Notes: When setting Max Battery Current, it must be set according to the battery specification.

Set a battery with maximum continuous output current 50A as an example:

single ESC, Max Battery Current: 50A

dual ESC, Max Battery Current: $50/2=25A$

4WD ESC, Max Battery Current: $50/4=12.5A$

The total output current of the battery must not exceed the battery maximum continuous output current specification. Otherwise, the BMS will cut off the power supply leading to motor's running out of control, which is very dangerous and may result in hurt for you. Moreover, please make sure that the Max Battery Current setting value does not exceed than that of the DC power supply to avoid ESC damage or unsafe happenings.

- **Max Battery Regen Current:** battery max continuous regenerative current (A)

Adjustable range: -500~0

Note: If using the DC power supply, the Max Battery Regen Current must be set to “0”. Otherwise, the large back EMF will damage the ESC or even the power supply when the motor brakes in a high speed mode.

6.2.4 Encoder Setup

The Encoder Setup includes ABI Encoder Direction, ABI Encoder Counts, Motor Pole Pairs and Encoder Offset as screenshot 6-11.

Screenshot 6-11 Encoder Setup

Encoder Setup			
ABI Encoder Direction	ABI Encoder Counts	Motor Pole Pairs	Encoder Offset(°)
Positive	4000	7	180

- **ABI Encoder Direction**

- ✧ Positive
- ✧ Negative

Parameter description: If the A and B phases are connected in reverse, the pulse count will be reversed. In this case, the encoder direction needs to be corrected.

- **ABI Encoder Counts**

Adjustable range: 0~100000

Parameter description: This parameter is 4 times the number of pulses output by the encoder when it rotates one circle. For example, if both phase A and phase B output 1000 pulses when they rotate one circle, they should be set to 4000 as 4 times of the frequency.

- **Motor Pole Pairs:** Number of pole pairs of the motor.

Adjustable range: 0~100

Parameter description: When the motor sensor port is set to encoder mode, this value will be calculated automatically after finishing the motor parameter identification. If no special application, no modification is required. Note: When starting in encoder mode, please ensure the correct setting of this value.

- **Encoder Offset(°):** The offset angle between the encoder and the motor rotor.

Adjustable range: 0~360°

Parameter Description: It means the offset angle between the encoder and the motor rotor. When the motor sensor port is set to encoder mode, this value will be calculated automatically after finishing the motor parameter identification. If no special application, no modification is required.

6.2.5 Speed PID Setup

The Speed PID Setup includes Speed PID Kp, Speed PID Ki, Speed PID Kd, PID Minimum Erpm, PID Maximum Erpm and PID Accel ($r/(m^2)$)Erpm as screenshot 6-12.

Screenshot 6-12 Speed PID Setup

Speed PID Setup					
Speed PID Kp	Speed PID Ki	Speed PID Kd	PID Minimum Erpm	PID Maximum Erpm	PID Accel($r/(m^2)$)Erpm
0.00380	0.00380	0.00010	500	100000	30000

- **Speed PID Kp:** Speed PID controller proportional gain coefficients.
Adjustable range: 0~10000
If no special application, no modification is required.
- **Speed PID Ki:** Speed PID controller integral gain coefficients.
Adjustable range: 0~10000
If no special application, no modification is required.
- **Speed PID Kd:** Speed PID controller differential gain coefficients.
Adjustable range: 0~10000
If no special application, no modification is required.
- **PID Minimum Erpm:** Minimum speed for speed PID control.
Adjustable range: 0~250000
Parameter description: When the ESC detects the Erpm lower than this min. value, it will stop output.
- **PID Maximum Erpm:** Maximum speed for speed PID control.
Adjustable range: 0~250000
Parameter description: When the ESC detects the Erpm higher than this max. value, it will maintain the maximum speed PID Maximum Erpm output.

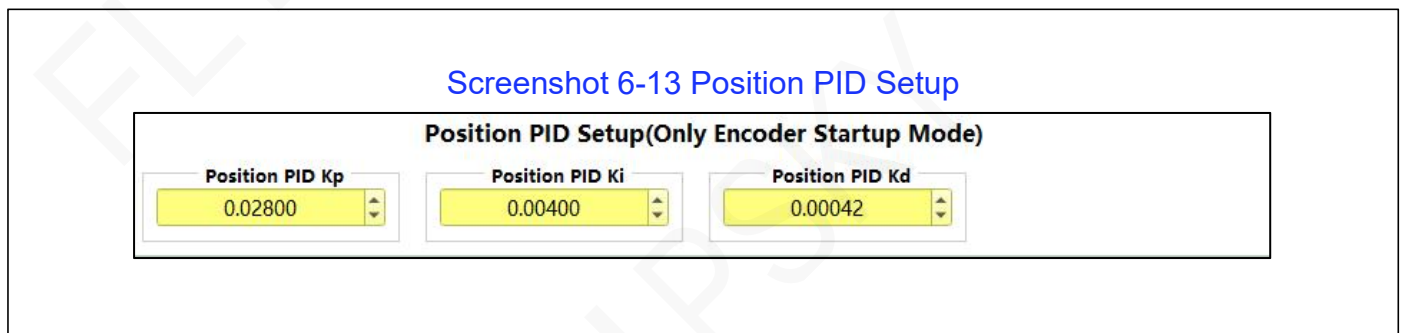
- **PID Accel ($r/(m^2)$)Erpm:**Acceleration value for speed PID control.

Adjustable range: -1~150000

Parameter description: The larger the value, the faster the speed changes. For example, when it is set to 30000 Erpm, it means that the speed changes by 30000 Erpm in 1 second. Which means that it takes 1 second only for the motor to increase Erpm from 0 to 30000 when a speed command of 30000 Erpm is given.

6.2.6 Position PID Setup

The Position PID Setup includes Position PID Kp, Position PID Ki and Position PID Kd as below screenshot 6-13.

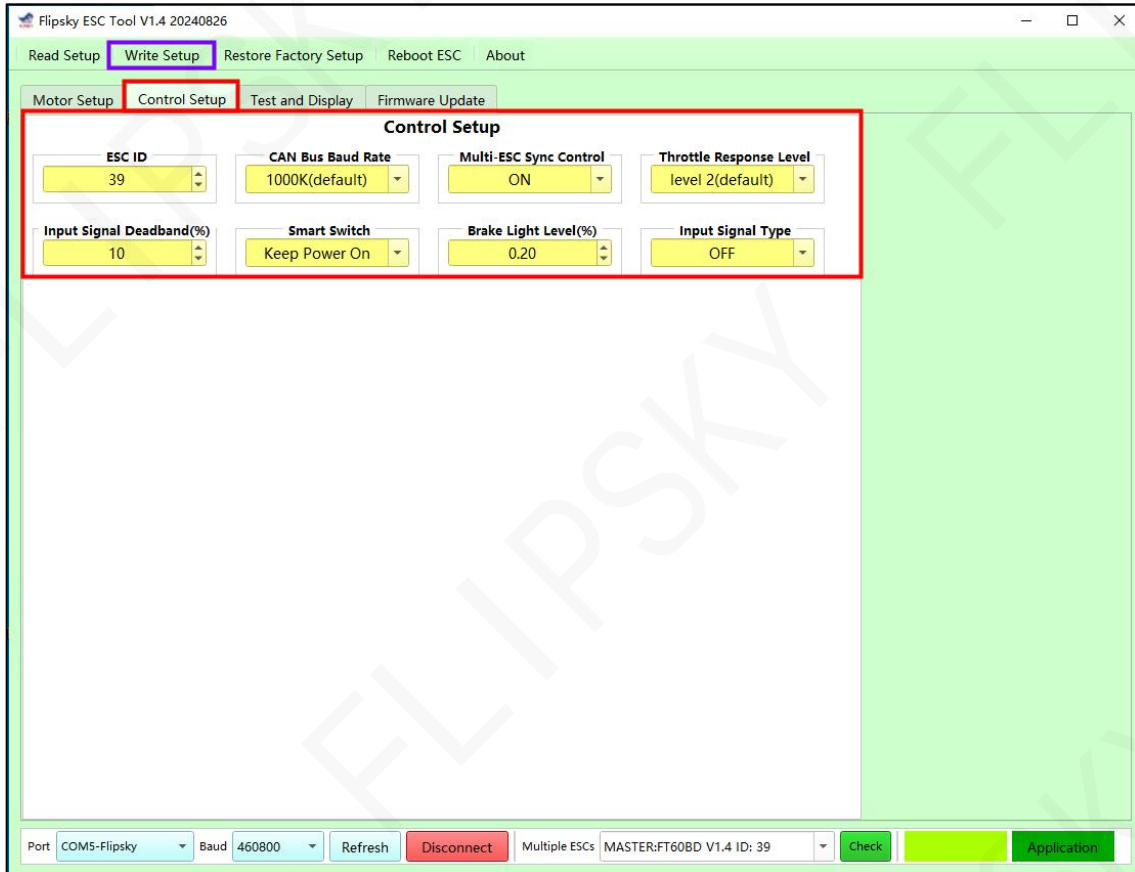


- **Position PID Kp:**Position PID controller proportional gain coefficients.
Adjustable range: 0~10000
If no special application, no modification is required.
- **Position PID Ki:**Position PID controller integral gain coefficients.
Adjustable range: 0~10000
If no special application, no modification is required.
- **Position PID Kd:**Position PID controller differential gain coefficients.
Adjustable range: 0~10000
If no special application, no modification is required.

7 Control Setup

The input control setup includes "ESC ID", "CAN Bus Baud Rate", "Multi-ESC Sync Control", "Throttle Response Level", "Input Signal Deadband", "Smart Switch", "Brake Light Level", "Input Signal Type" as below Screenshot 7-1.

Screenshot 7-1.Control Setup Interface

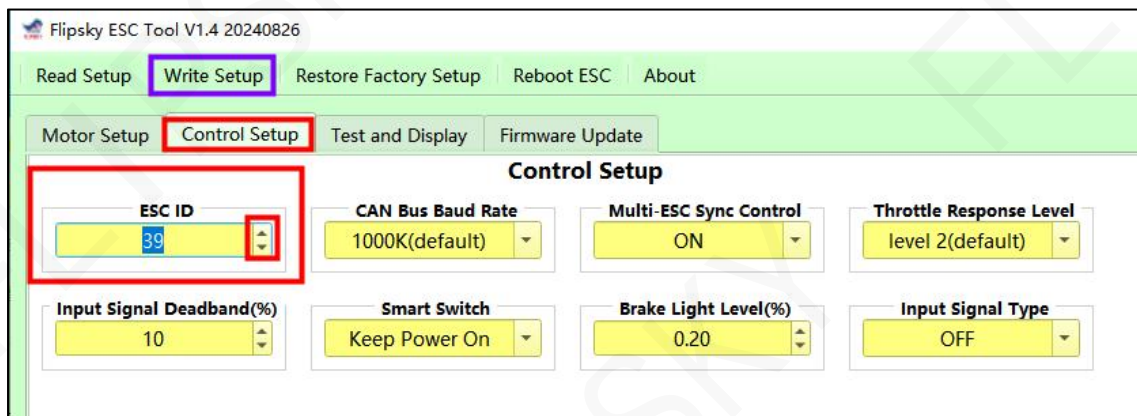


7.1 ESC ID Setup

ESC ID: ESC ID Number (See screenshot 7-2) adjustable range: 1~254 for multiple ESCs via CAN. Please make sure the uniqueness of the ESC ID since the same ESC IDs result in parameter identification failures or abnormal control. ESC ID 0 (special command number) and ESC ID 255 (special broadcast command) are not available for setup.

ESC ID number setup as Screenshot 4.2.1.

Screenshot 7-2.ESC ID

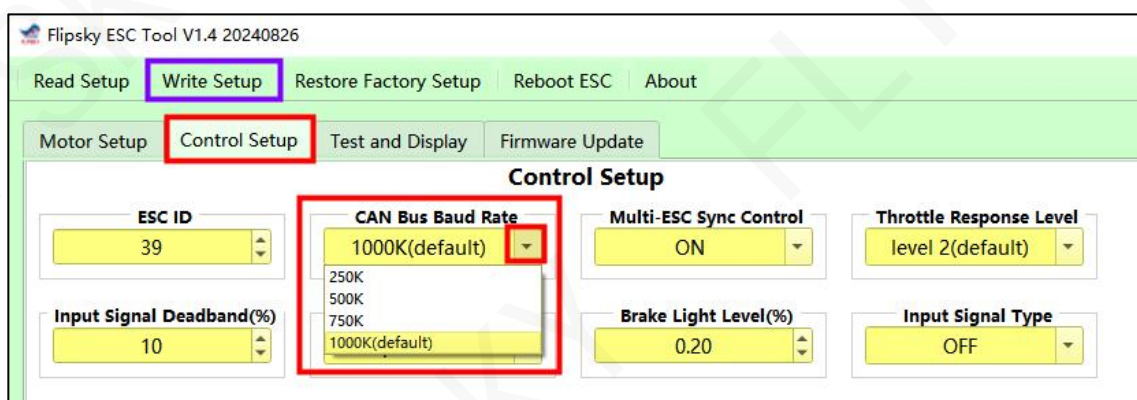


7.2 CAN-Bus Baud Rate Setup

CAN-Bus Baud Rate (see as Screenshot 7-3): Please ensure the consistency of CAN baud rate for the same CAN devices.

CAN-Bus baud rate setup as 4.2.2.

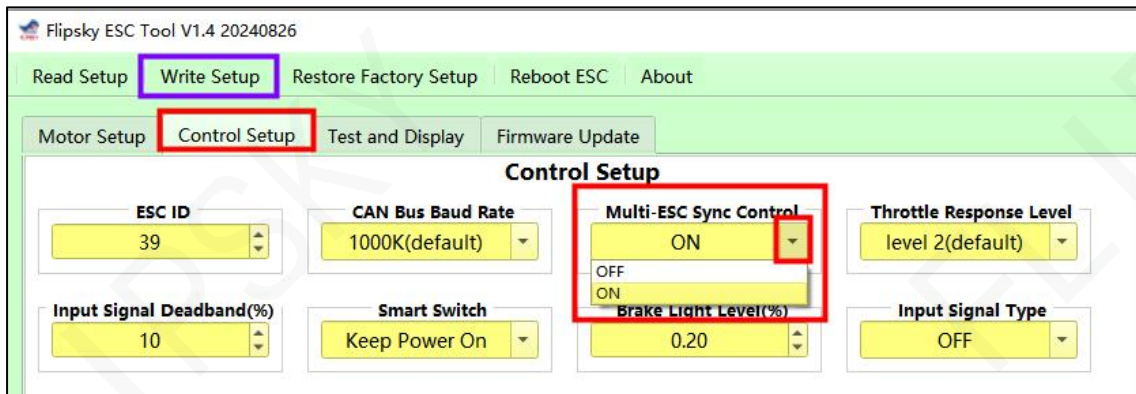
Screenshot 7-3 CAN-Bus Baud Rate



7.3 Multiple ESCs Sync Control Setup

Multi-ESC Sync Control: synchronous control for multiple ESCs via CAN as screenshot 7-4.

Screenshot 7-4.CAN Multiple ESCs



Setting steps as below:

- Click "Control Setup" to enter the setup interface.
- Click "Multi-ESC Sync Control" to select whether synchronous control of the CAN ESCs or not. (ON means Synchronous Control, OFF means not)
- Click the upper left menu "Write Setup" to save the setup

Parameter Description:

When the ESC is the master board and the "Multi-ESC Sync Control" is set to "ON", the input signal will synchronously control the master ESC and all CAN slave ESCs. If the "Multi-ESC Sync Control" is set to "OFF" in this situation, then the input signal controls the master ESC only.

When the ESC is the slave board and the "Multi-ESC Sync Control" is set to "ON", this slave ESC will under the control of master CAN-Bus. If the "Multi-ESC Sync Control" is set to "OFF" in this case, the slave ESC will shield the control commands from the master CAN-Bus.

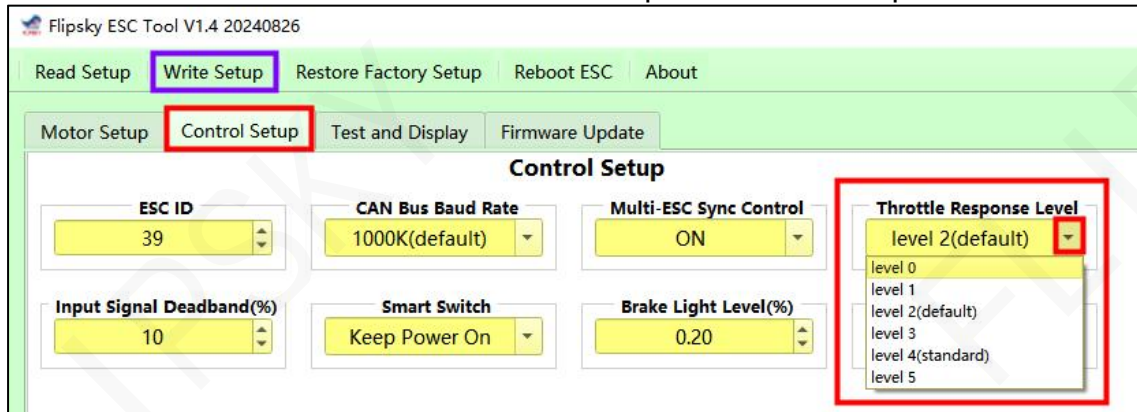
Therefore, if you want multiple ESCs to be controlled synchronously, set all ESC's Multi-ESC Sync Control "ON". However, if you want to control each ESC separately in UART, PPM or ADC control type, set all ESC's Multi-ESC Sync Control "OFF".

If use your own control module as the master board and want to control the ESCs via CAN-Bus, set all ESC's Multi-ESC Sync Control "ON", and the Input Signal Type "OFF".

7.4 Throttle Response Level Setup

Throttle Response Level: Level 0-Level 5 (6 level choices as screenshot 7-5).

Screenshot 7-5 Throttle Response Level Setup



Setting steps as below:

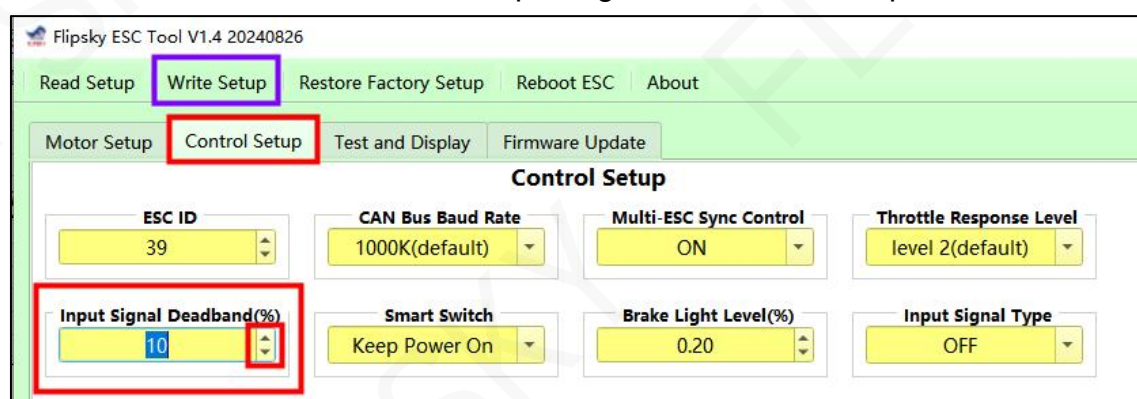
- Click "Control Setup" to enter the setup interface.
- Click "Throttle Response Level" to select the appropriate throttle response level.
- Click the upper left menu "Write Setup" to save the setup.

Parameter description: The higher of the level, the more sensitive of the throttle and brake response. The default level is Level 2, and Level 4 is the standard level which has no lead or lag on all input signals and is suitable for customers customized requirement.

7.5 Input Deadband Setup

Input Signal Deadband: Adjustable range: 1%-25% as below screenshot 7-6:

Screenshot 7-6. Input Signal Deadband Setup



Setting steps as below:

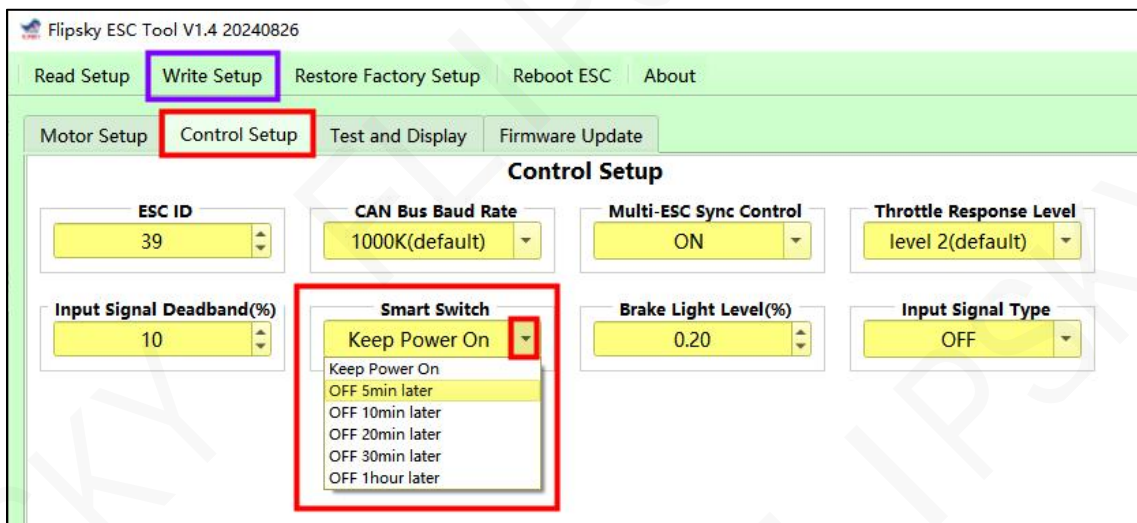
- Click "Control Setup" to enter the setup interface.
- Click "Input Signal Deadband" to adjust the appropriate percentage;
- Click the upper left menu "Write Setup" to save the setup.

Parameter description: The ESC starts the current output only when the throttle input signal percentage is over than that of the input deadband setup . It's recommended to use the default value 10%. Regarding the input control type PPM and UART, the deadband percentage can be small or even 0% as the remote controller has set already, for example, the Flipsky remote controller VX4 with 5% deadband already . It's also fine to keep the default setting 10% as the ADC control type recommended.

7.6 Smart Switch

Smart Switch: support automatic shutdown mode setting. This function is applicable for FT ESC FT85BD and FT60BD only. It can be set through Smart Switch as shown in screenshot 7-7:

Screenshot 7-7 Smart Switch Setup

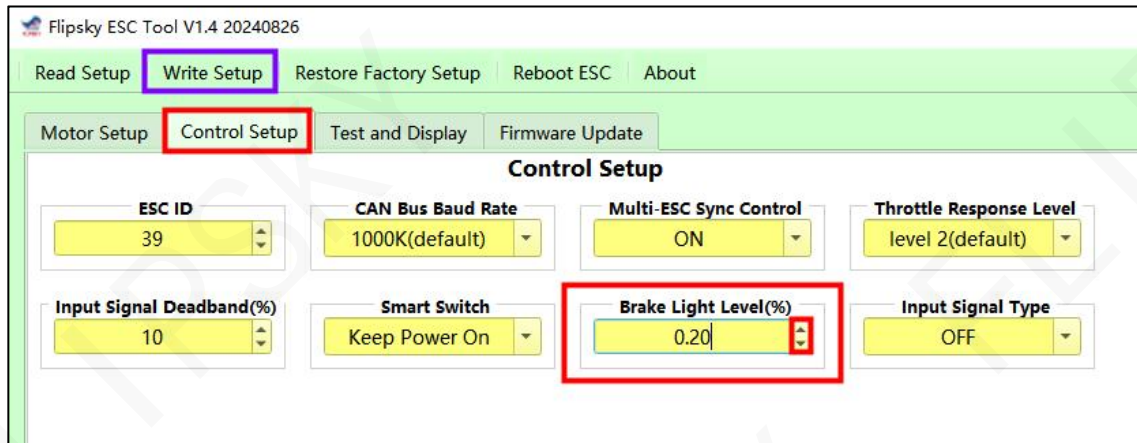


Parameter description: "Keep Power On" standards for always on without poweroff. "OFF 5min later" supports auto shutoff without driving signal after 5 minutes later.

7.7 Brake Light Level Setup

Brake Light Level: The brake light brightness setup in static.

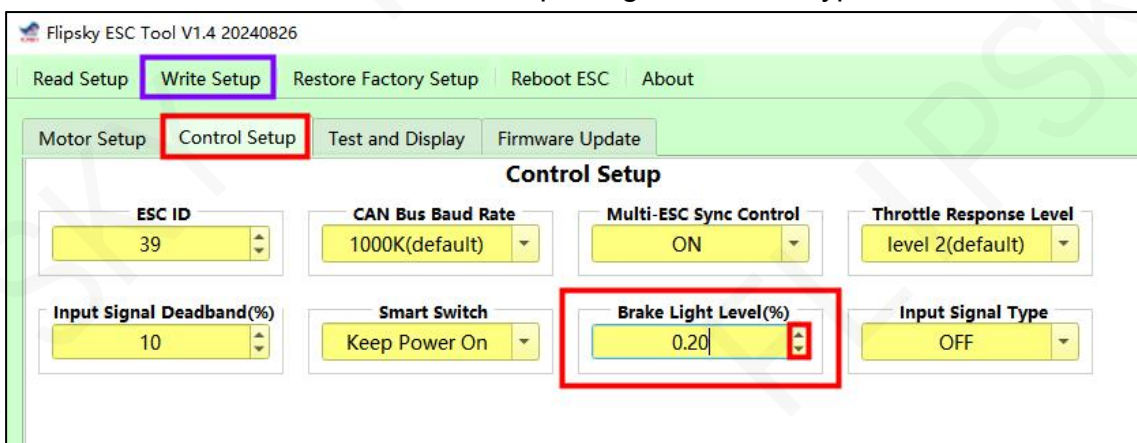
Screenshot 7-8 Brake Light Level Setup



7.8 Input Control Type Setup

The input signal control types: OFF, UART, PPM, ADC and PAS (5choices) and the default setting is OFF as below screenshot 7-9.

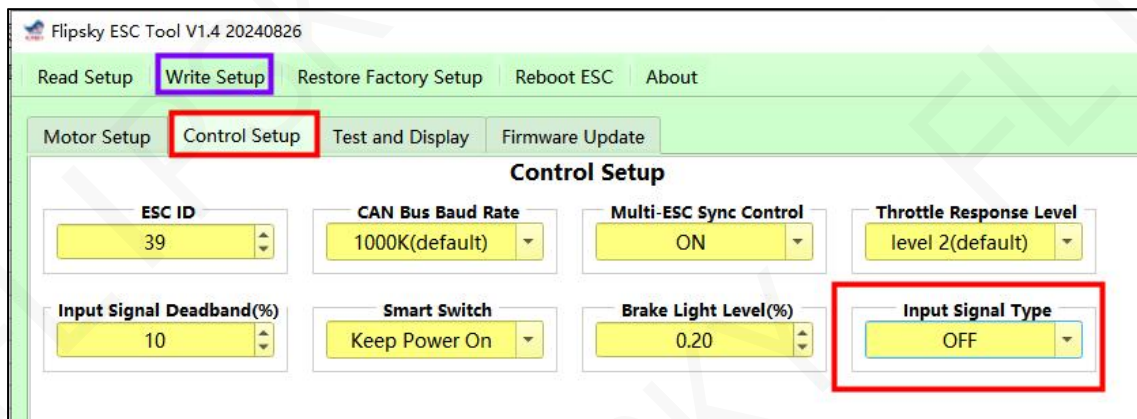
Screenshot 7-9. Input Signal Control Type



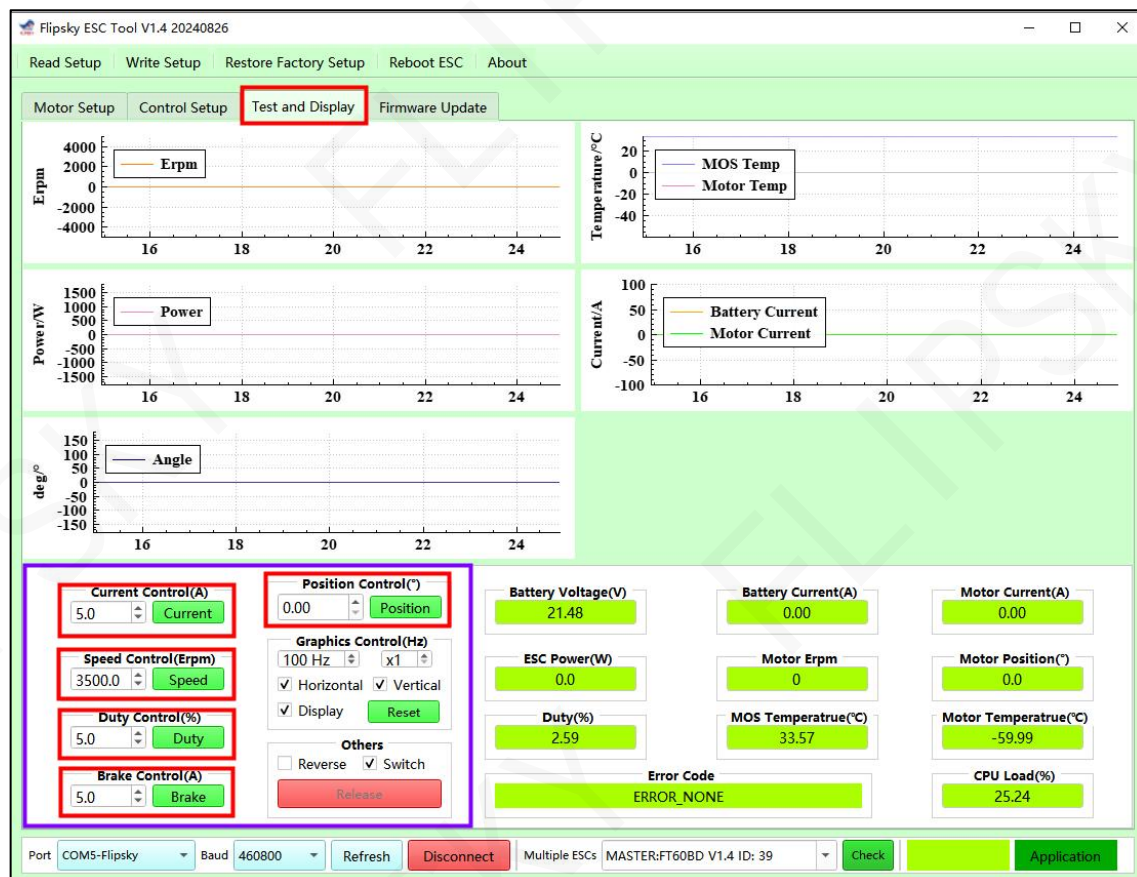
7.8.1 OFF

If the "Input Signal Type" is set to "OFF" (Screenshot 7-10), it means that all input signals control will be shielded. The ESC can be controlled through the Test and Display interface button control only like "Current Control", "Duty Control" and "Brake Control" (Screenshot 7-11). Please refer to 8.1.1 for button control setup.

Screenshot 7-10 Input Signal Type OFF



Screenshot 7-11. Test and Display Interface



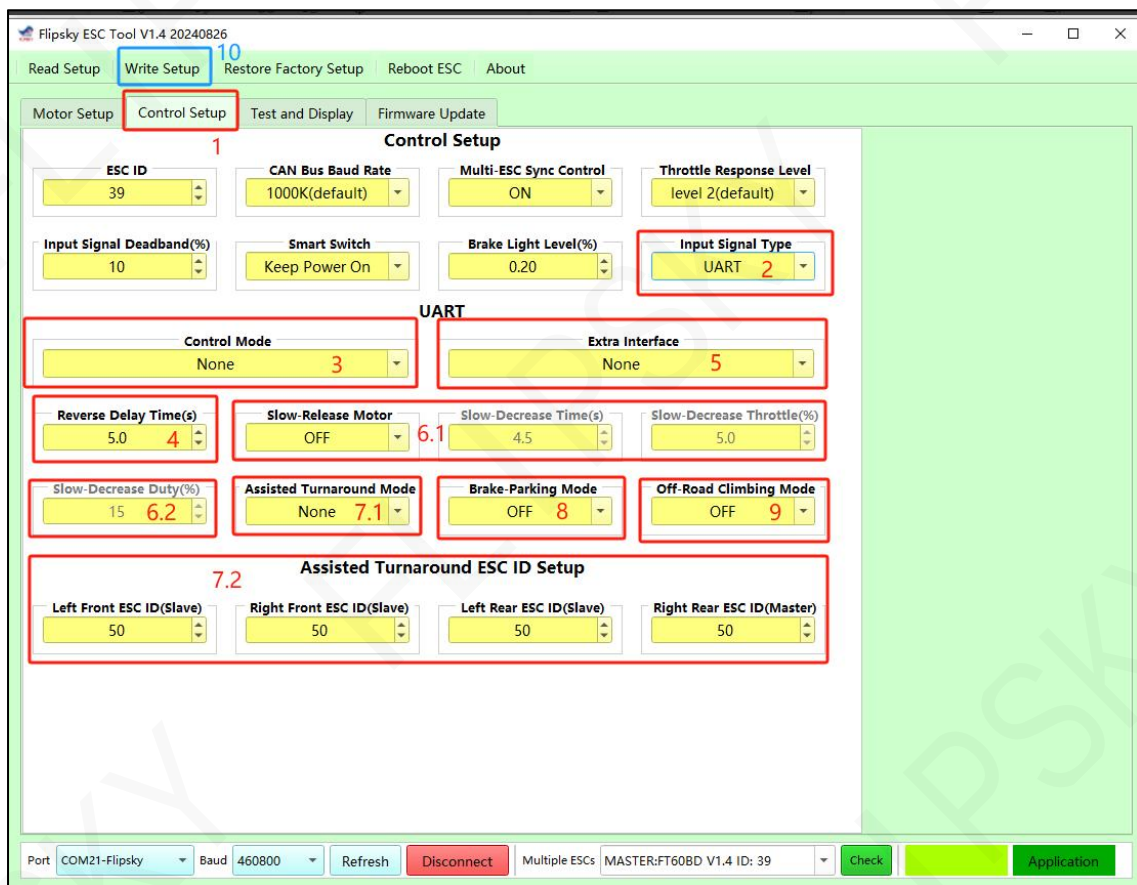
7.8.2 UART

7.8.2.1 UART Description and Setup

The UART mode supports Flipsky remote controller VX4's latest firmware VX4_V1_2 and remote controller VX3 pro only. The max EPRM can be set under different gear speed levels for this mode.

The UART setup interface as below screenshot 7-12.

Screenshot 7-12 UART Setup Interface



Note: step 3 and step 10 are “must” steps to be done, and other parameters can be set according to your needs.

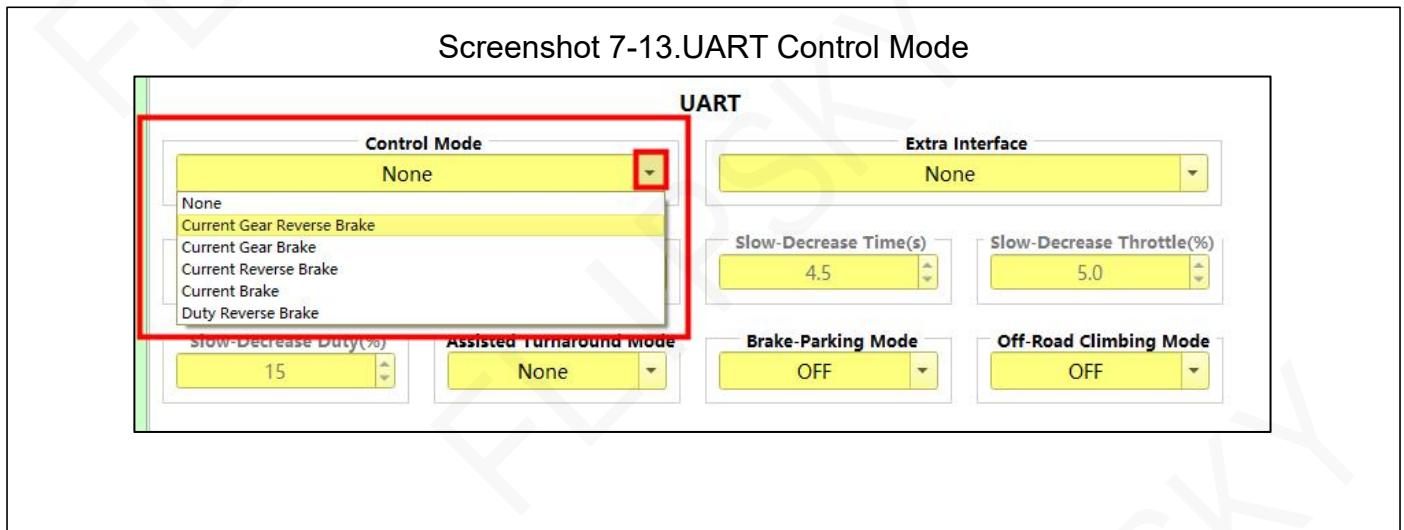
Setting steps as below:

- Step 1: click "Control Setup" to enter the control setup interface
- Step 2: select UART for "Input Signal Type"
- Step 3: select the needed UART Control Mode (see detailed description in 7.8.2.2)
- Step 4: set UART Reverse Delay Time (see detailed description in 7.8.2.3)

- Step 5: set Extra Interface according to needs (see detailed description in 7.8.2.4)
- Step 6: set Slow-Release Motor (see detailed description in 7.8.2.5)
- Step 7: set Assisted Turnaround Mode and Assisted Turnaround ESC ID Setup according to needs (see detailed description in 7.8.2.6)
- Step 8: set Brake Parking Mode (see detailed description in 7.8.2.7)
- Step 9: set Off-Road Climbing Mode (see detailed description in 7.8.2.8)
- Step 10: Click "Write Setup" to save all parameters settings.

7.8.2.2 UART Control Mode Setup

Select the control mode per your application needs as below screenshot 7-13.



- None
- Current Gear Reverse Brake: Current control mode: gear speed level limit, reverse with brake, reverse
- Current Gear Brake: Current control mode: gear speed level limit, reverse with brake
- Current Reverse Brake: Current control mode: reverse with brake, reverse
- Current Brake: Current control mode: reverse with brake
- Duty Gear Brake Reverse: Dutycycle control mode: gear speed level limit, reverse with brake, reverse

7.8.2.3 UART Reverse Delay Time Setup

The UART control mode supports reverse function, so you can set the UART Reverse Delay Time per your needs. The adjustable range: 0.0s~10s, and the default is 5s as shown in 7-14:

Screenshot 7-14 UART Reverse Delay Time Setup

UART

Control Mode None	Extra Interface None
Reverse Delay Time(s) 5.0	Slow-Release Motor OFF
Slow-Decrease Time(s) 4.5	Slow-Decrease Throttle(%) 5.0
Slow-Decrease Duty(%) 15	Assisted Turnaround Mode None
Brake-Parking Mode OFF	Off-Road Climbing Mode OFF

Parameter description: When the motor is stopped, maximum pull the throttle back and start to rotate in the reverse direction. The maximum speed of the reverse can reach within the set time period (as shown in 7-14, it's 5 seconds as this set).

7.8.2.4 UART Extra Interface Setup

The Extra Interface is applicable to Flipsky ESC FT85BD only which supports the functions of front light, rear brake light and horn in UART control mode. The remote VX4's UART mode is available at present.

Do Extra Interface settings per your needs as below screenshot 7-15.

Screenshot 7-15.Extra Interface

UART

Control Mode None	Extra Interface None
Reverse Delay Time(s) 5.0	Slow-Release Motor OFF
Slow-Decrease Duty(%) 15	Assisted Turnaround Mode None

- None (No extra interface connections)

- Front Light
- Brake Light
- Horn
- Front and Brake Light
- Front and Horn
- Brake Light and Horn
- Front, Brake Light and Horn

7.8.2.5 UART Slow-Release Setup

Slow-Release Motor function enables FT series ESCs to maintain a certain current output after releasing the throttle under the UART input signal, which can offset the resistances of high gear ratios (>3.5:1) off-road skateboards or other sports kits and put the skateboards or sports kits in a release state with almost zero resistance and smoother starting process.

The Slow-Release setup per your needs as below screenshot 7-16.

Screenshot 7-16.Slow-Release Motor

UART			
Control Mode None	Extra Interface None		
Reverse Delay Time(s) 5.0	Slow-Release Motor OFF	Slow-Decrease Time(s) 4.5	Slow-Decrease Throttle(%) 5.0
Slow-Decrease Duty(%) 15	Assisted Turnaround Mode None	Brake-Parking Mode OFF	Off-Road Climbing Mode OFF

Parameters Description:

- **Slow-Release Motor:** Setup column, default: OFF
- **Slow-Decrease Time:** current output releasing time, default: 4.5s
- **Slow-Decrease Throttle:** current output throttle percentage, default:5% (0.05×Motor Current)
- **Slow-Decrease Duty:** the motor duty cycle percentage critical point

Notes: This function is not available under the control mode of Duty Gear Brake Reverse.

7.8.2.6 Assisted Turnaround ESC ID Setup

Assisted Turnaround Mode Setup: This assisted turnaround function is applicable in UART control only. Also, it is only supported by the latest firmware VX4_V1_4 or above of VX4 remote controller. Please review more detailed description via the VX4 and VX4Pro user manual.

Steps of Assisted Turnaround Mode setup:

Step 1: Select the Assisted Turnaround Mode per your building as 7-17.

Screenshot 7-17 Assisted Turnaround Mode Setup

The screenshot displays the UART control interface with various settings. The 'Assisted Turnaround Mode' dropdown menu is highlighted with a red box, showing the following options: None, Dual Drive Mode, and Quad Drive Mode. Other settings visible include Control Mode (None), Extra Interface (None), Reverse Delay Time(s) (5.0), Slow-Release Motor (OFF), Slow-Decrease Time(s) (4.5), Slow-Decrease Throttle(%) (5.0), Slow-Decrease Duty(%) (15), Brake-Parking Mode (OFF), Off-Road Climbing Mode (OFF), Left Front ESC ID(Slave) (50), Right Front ESC ID(Slave) (50), Left Rear ESC ID(Slave) (50), and Right Rear ESC ID(Master) (50).

Parameter description:

- Assisted Turnaround Mode
- None: OFF the Assisted Turnaround function
- Dual Drive Mode: for two wheels driving. Two motors run in different direction during the turnaround process.
- Quad Drive Mode: for four wheels driving. One side motors rotate forward, but the other side motors rotate different direction reversely during the turnaround process.

Step 2: Do Assisted Turnaround ESC ID Setup as 7-18.

Screenshot 7-18. Assisted Turnaround ESC ID Setup

The screenshot displays the 'Assisted Turnaround ESC ID Setup' interface. A red box highlights the four ESC ID input fields, which are all set to 50. The fields are labeled: Left Front ESC ID(Slave), Right Front ESC ID(Slave), Left Rear ESC ID(Slave), and Right Rear ESC ID(Master).

- Assisted Turnaround ESC ID Setup
- Left Front ESC ID(Slave)
- Right Front ESC ID(Slave)
- Left Rear ESC ID(Slave)
- Right Rear ESC ID(Master)
- Dual Drive Mode: Setup "Left Rear ESC ID(Slave)" and "Right Rear ESC ID(Master)"
- Quad Drive Mode: Setup "Left Front ESC ID(Slave)", "Right Front ESC ID(Slave)", "Left Rear ESC ID(Slave)", "Right Rear ESC ID(Master)".

7.8.2.7 Brake-Parking Mode Setup

Brake-Parking Mode: Pull the remote controller throttle gear/bar 0.5S to enter into Brake-Parking Mode after the motor stop running. The greater percentage of pulling back the throttle gear, the bigger of the parking force. Setup as shown in 7-19:

Screenshot 7-19 Brake-Parking Mode Setup

UART			
Control Mode None		Extra Interface None	
Reverse Delay Time(s) 5.0	Slow-Release Motor OFF	Slow-Decrease Time(s) 4.5	Slow-Decrease Throttle(%) 5.0
Slow-Decrease Duty(%) 15	Assisted Turnaround Mode None	Brake-Parking Mode OFF OFF ON	Off-Road Climbing Mode OFF

Assisted Turnaround ESC ID Setup

Note: This mode injects current into the motor d-axis to make the coil generate magnetic force and attract the rotor to achieve the parking effect, so current will be consumed when parking. The maximum d-axis current can reach Motor Current Max when parking. It is not recommended to park in this mode for a long time, otherwise the motor will get hot and lead to overload current limiting phenomenon.

Note: When in Reverse Mode like Current Gear Reverse Brake or Current Gear Brake mode, max pulling the throttle gear / bar will end the Brake-Parking Mode but enter into Reverse Mode instead.

At present, this mode supports for remote VX4 (firmware version: V1.4) and VX4 Pro (firmware version: V1.0) only. For more detailed description, please refer to the VX4 and VX4 Pro user manuals.

7.8.2.8 Off-Road Climbing Mode Setup

Off-Road Climbing Mode: In this mode, all CAN escs will be switched to duty cycle control and allow users to pass off-road passing gravel roads or climbing high slops easily by controlling speed of the four wheels synchronously via remote controller. Setup as shown in 7-20

Screenshot 7-20 Off-Road Climbing Mode Setup

UART

Control Mode: None

Extra Interface: None

Reverse Delay Time(s): 5.0

Slow-Release Motor: OFF

Slow-Decrease Time(s): 4.5

Slow-Decrease Throttle(%): 5.0

Slow-Decrease Duty(%): 15

Assisted Turnaround Mode: None

Brake-Parking Mode: OFF

Off-Road Climbing Mode: OFF

Assisted Turnaround ESC ID Setup

Note:

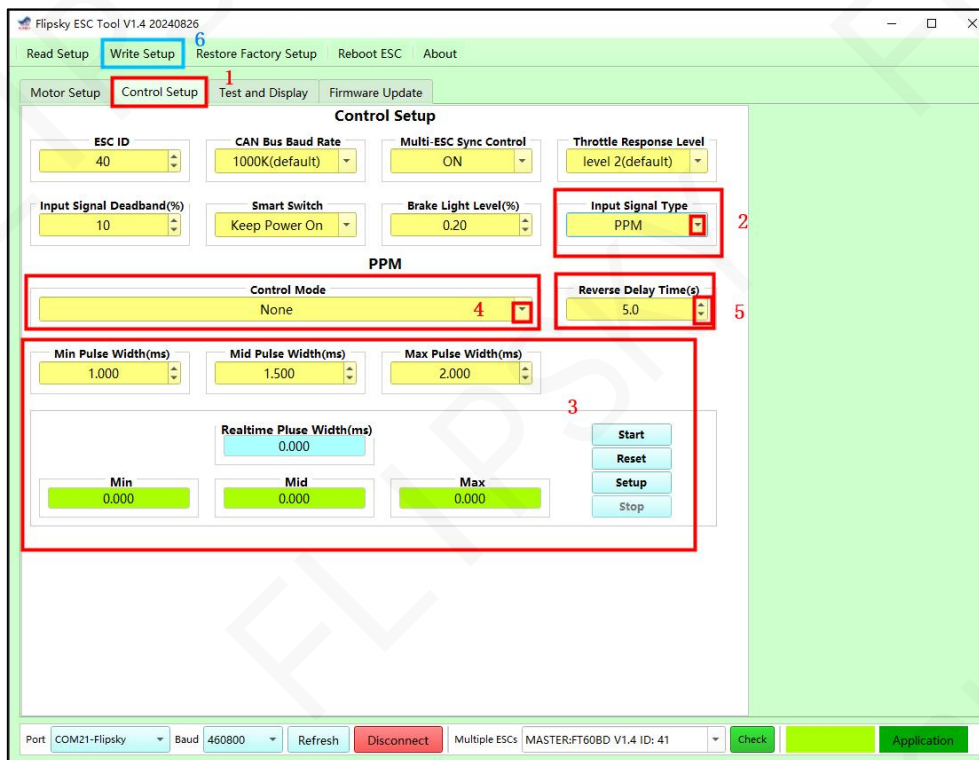
1. After using this mode, the reverse mode will be disabled.
2. At present, this mode supports for remote VX4 (firmware version: V1.4) and VX4 Pro (firmware version: V1.0) only. For more detailed description, please refer to the VX4 and VX4 Pro user manuals.

7.8.3 PPM

7.8.3.1 PPM Description and Setup

The PPM signal pulse width usually ranges from 1ms to 2ms, so theoretically the PPM control frequency supports up to 500Hz. Considering the MCU clock accuracy, it is recommended to use the PPM control frequency less than 400Hz. Setting steps as below screenshot 7-21.

Screenshot 7-21 PPM Input Setup



Setting steps as below:

- Step 1: Click "Control Setup" to enter the setup interface.
- Step 2: Switch the "Input Signal Type" to "PPM".
- Step 3: Do PPM throttle calibration for the first time as details in 7.8.3.2.

(Note: Doing PPM throttle calibration for one side of the dual esc controller is workable. The other side's Input Signal Type set "OFF".)

- Step 4: Select the appropriate PPM Control Mode per your needs (see details in 7.8.3.3).
- Step 5: Set the Reverse Delay Time (see details in 7.8.3.4).
- Step 6: Click the upper left menu "Write Setup" to save the setup.

7.8.3.2 PPM Input Throttle Calibration

Please do PPM input setup for the first time application. Setting Flipsky Remote VX1 as an example:

Click "Start" to begin the calibration as below screenshot 7-22.

Screenshot 7-22 Start PPM Input Setup

Min Pulse Width(ms)	Mid Pulse Width(ms)	Max Pulse Width(ms)
1.000	1.500	2.000

Realtime Pulse Width(ms)
1.497

Min	Mid	Max
1.496	1.497	1.499

Start
Reset
Setup
Stop

Notes: No PPM signal can be detected if the remote receiver hasn't been plugged with the ESC PPM wire when you click "Start" as below screenshot 7-23.

Screenshot 7-23. No PPM Signal Data

Min Pulse Width(ms)	Mid Pulse Width(ms)	Max Pulse Width(ms)
0.000	0.000	0.000

Realtime Pulse Width(ms)
0.000

Min	Mid	Max
0.000	0.000	0.000

Start
Reset
Setup
Stop

Connect the remote receiver with the ESC PPM wire in this case. If the PPM signal data seems abnormal (as shown in Screenshot 7-24), please click "Reset". (Refer to Screenshot 7-22).

Screenshot 7-24. PPM Signal Data Abnormal

Min Pulse Width(ms)	Mid Pulse Width(ms)	Max Pulse Width(ms)
0.000	0.749	1.498

Realtime Pulse Width(ms)
1.497

Min	Mid	Max
0.000	0.749	1.498

Start
Reset
Setup
Stop

Max pull and push the remote throttle gear to refresh the minimum, middle and maximum PPM signal values as below screenshot 7-25 and screenshot 7-26.

Screenshot 7-25. PPM signal max.

Min Pulse Width(ms)	Mid Pulse Width(ms)	Max Pulse Width(ms)
1.000	1.500	2.000

Realtime Pluse Width(ms)
1.992

Min	Mid	Max
1.496	1.746	1.995

Start
Reset
Setup
Stop

Screenshot 7-26. PPM signal min.

Min Pulse Width(ms)	Mid Pulse Width(ms)	Max Pulse Width(ms)
1.000	1.500	2.000

Realtime Pluse Width(ms)
1.000

Min	Mid	Max
0.999	1.497	1.995

Start
Reset
Setup
Stop

Click "Setup" to write the "Min", "Mid" and "Max" data to Pulse Width accordingly as screenshot 7-27.

Screenshot 7-27. Set PPM Pulse Width Values

Min Pulse Width(ms)	Mid Pulse Width(ms)	Max Pulse Width(ms)
0.999	1.497	1.995

Realtime Pluse Width(ms)
1.497

Min	Mid	Max
0.999	1.497	1.995

Start
Reset
Setup
Stop

Click "Stop" to complete the PPM input setup as screenshot 7-28.

Screenshot 7-28.PPM Input Setup Finish

The screenshot displays the 'PPM' configuration window. At the top, 'Control Mode' is set to 'Current Reverse(reverse:1.0~1.5ms,forward:1.5~2.0ms)'. To the right, 'Reverse Delay Time(s)' is set to 0.5. Below these, three pulse width settings are shown: 'Min Pulse Width(ms)' at 1.000, 'Mid Pulse Width(ms)' at 1.499, and 'Max Pulse Width(ms)' at 1.994. A 'Realtime Pluse Width(ms)' section shows 1.499. At the bottom, 'Min', 'Mid', and 'Max' values are displayed in green boxes as 1.000, 1.499, and 1.994 respectively. On the right side, there are four buttons: 'Start', 'Reset', 'Setup', and 'Stop'. The 'Stop' button is highlighted with a red rectangular box.

7.8.3.3 PPM Control Mode Setup

Select the appropriate PPM Control Mode per your needs as screenshot 7-29:

Screenshot 7-29.PPM Control Mode Setup

The screenshot shows the 'Control Mode' dropdown menu open. The menu lists the following options: 'None', 'Current(forward:1.0~2.0ms)', 'Current Brake(brake:1.0~1.5ms,forward:1.5~2.0ms)', 'Current Reverse(reverse:1.0~1.5ms,forward:1.5~2.0ms)', 'Current Reverse Brake(reverse/brake:1.0~1.5ms,forward:1.5~2.0ms)', 'Duty(forward:1.0~2.0ms)', 'Duty Reverse(reverse:1.0~1.5ms,forward:1.5~2.0ms)', and 'Duty Reverse Brake(reverse/brake:1.0~1.5ms,forward:1.5~2.0ms)'. The 'Current Reverse' option is highlighted in yellow. The dropdown arrow icon is circled in red.

- None
- Current (forward:1.0~2.0ms)
Current control mode, unidirectional control, only forward:1.0~2.0ms.
- Current Brake (brake:1.0~1.5ms,forward:1.5~2.0ms)
Current control mode, bidirectional control, brake:1.0~1.5ms, forward:1.0~2.0ms.
- Current Reverse (reverse1.0~1.5ms, forward:1.5~2.0ms)

Current control mode, bidirectional control, no brake, backward:1.0~1.5ms, forward:1.0~2.0ms.

- Current Reverse Brake(reverse/brake:1.0~1.5ms, forward:1.5~2.0ms)

Current control mode, bidirectional control, brake/reverse:1.0~1.5ms, forward:1.0~2.0ms.

- Duty(forward:1.0~2.0ms)

Duty control mode, unidirectional control, only forward:1.0~2.0ms.

- Duty Reverse(reverse:1.0~1.5ms, forward:1.5~2.0ms)

Duty cycle control mode, bidirectional control, backward:1.0~1.5ms, forward:1.0~2.0ms.

- Duty Reverse Brake(reverse/brake:1.0~1.5ms, forward:1.5~2.0ms)

Duty cycle control mode, bidirectional control, brake/reverse:1.0~1.5ms, forward:1.0~2.0ms.

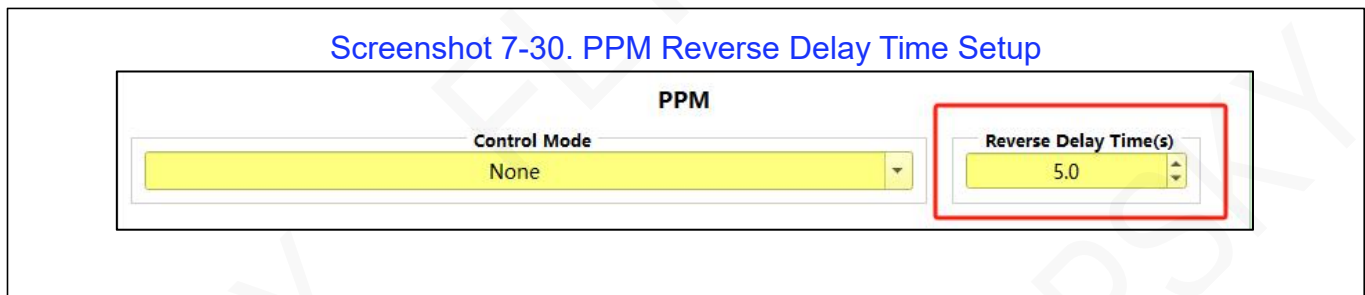
- Duty Reverse Slow(reverse:1.0~1.5ms, forward:1.5~2.0ms)

Duty cycle control mode, bidirectional control, reverse:1.0~1.5ms, forward:1.0~2.0ms.

Note: No back EMF is generated when the accelerator is released.

7.8.3.4 PPM Reverse Delay Time Setup

The PPM control mode supports reverse function, so you can set the UART Reverse Delay Time per your needs. The adjustable range: 0.0s~10s, and the default is 5s as shown in 7-30:



Parameter description: When the motor is stopped, maximum pull the throttle back and start to rotate in the reverse direction. The maximum speed of the reverse can reach within the set time period (as shown in 7-30, it's 5 seconds as this set).

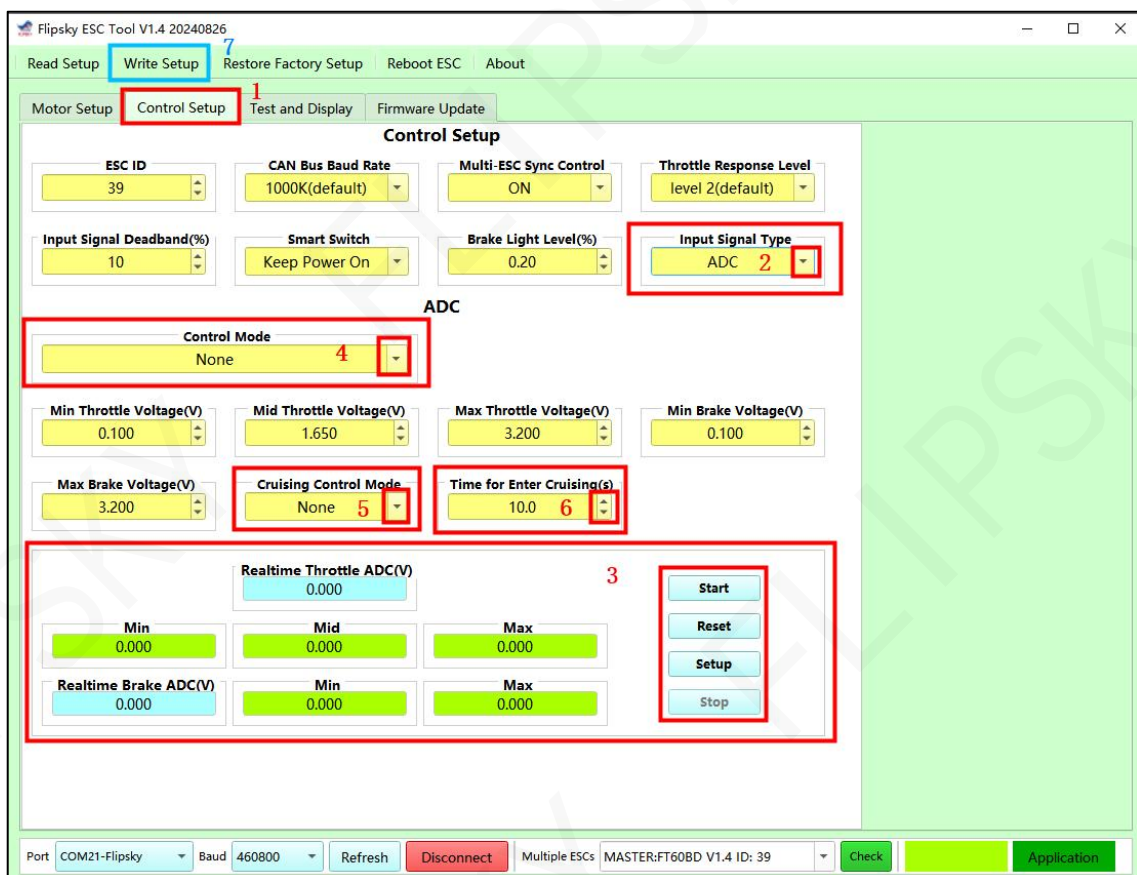
7.8.4 ADC

7.8.4.1 ADC Description and Setup

ADC control type voltage range: 0V~3.3V. Please make sure the ADC input voltage not over than 3.3V to avoid the ESC 3.3V chip damage. Setting steps as below screenshot 7-31.

- Step 1: Click "Control Setup" to enter the setup interface.
- Step 2: Switch the "Input Signal Type" to "ADC" (Note: Doing ADC throttle calibration for one side of the dual esc controller is workable. The other side's Input Signal Type set "OFF").
- Step 3: Do ADC throttle calibration for the first time as details in 7.8.4.2.
- Step 4: Select the appropriate ADC Control Mode per your needs (see details in 7.8.4.3).
- Step 5: Select the Cruising Control Mode per your needs (refer to details in 7.8.4.4)
- Step 6: Set the Time for Enter Cruising (refer to details in 7.8.4.4)
- Step 7: Click the upper left menu "Write Setup" to save the setup.

Screenshot 7-31. ADC Control Type Setup



7.8.4.2 ADC Input Throttle Calibration

Do ADC throttle calibration for the first time as follows:

Click "Start" to begin the calibration as below screenshot 7-32.

Screenshot 7-32.ADC Input Setup

The screenshot shows the 'ADC Input Setup' interface. It features two rows of input fields. The top row is for 'Realtime Throttle ADC(V)' with values: Min (0.863), Mid (0.865), and Max (0.866). The bottom row is for 'Realtime Brake ADC(V)' with values: Min (0.836), Mid (0.836), and Max (0.839). To the right of these fields are four buttons: 'Start' (highlighted with a red box), 'Reset', 'Setup', and 'Stop'.

Realtime Throttle ADC(V)		
Min	Mid	Max
0.863	0.865	0.866

Realtime Brake ADC(V)		
Min	Mid	Max
0.836	0.836	0.839

Buttons: Start, Reset, Setup, Stop

Max twisting the ADC throttle to flash the min and max voltages as below screenshot 7-33.

Screenshot 7-33.Flash the ADC Throttle Forward

The screenshot shows the 'ADC Input Setup' interface after flashing the throttle. The 'Realtime Throttle ADC(V)' row now shows: Min (0.861), Mid (1.707), and Max (2.552). The 'Realtime Brake ADC(V)' row remains the same: Min (0.836), Mid (0.832), and Max (0.840). The 'Start' button is still highlighted with a red box.

Realtime Throttle ADC(V)		
Min	Mid	Max
0.861	1.707	2.552

Realtime Brake ADC(V)		
Min	Mid	Max
0.836	0.832	0.840

Buttons: Start, Reset, Setup, Stop

Max press or pull the brake to flash the min and max voltages as below screenshot 7-34.

Screenshot 7-34. Flash the ADC Brake

The screenshot shows the 'ADC Input Setup' interface after flashing the brake. The 'Realtime Throttle ADC(V)' row remains the same: Min (0.861), Mid (1.707), and Max (2.552). The 'Realtime Brake ADC(V)' row now shows: Min (0.832) and Max (2.554). The 'Start' button is still highlighted with a red box.

Realtime Throttle ADC(V)		
Min	Mid	Max
0.861	1.707	2.552

Realtime Brake ADC(V)		
Min	Max	
0.832	2.554	

Buttons: Start, Reset, Setup, Stop

Click "Setup" to write the "Min" , "Mid" and "Max" data to throttle and brake voltage accordingly as screenshot 7-35.

Screenshot 7-35. ADC Real-time Throttle and Brake Voltage Setup

The screenshot displays the 'ADC Real-time Throttle and Brake Voltage Setup' window. It features several input fields for voltage settings, a real-time ADC display, and a set of control buttons. The 'Setup' button is highlighted with a red box.

Parameter	Value
Min Throttle Voltage(V)	0.100
Mid Throttle Voltage(V)	1.650
Max Throttle Voltage(V)	3.200
Min Brake Voltage(V)	0.100
Max Brake Voltage(V)	3.200
Cruising Control Mode	None
Time for Enter Cruising(s)	10.0
Realtime Throttle ADC(V)	0.865
Realtime Brake ADC(V)	0.835
Min (Throttle)	0.860
Mid (Throttle)	1.706
Max (Throttle)	2.552
Min (Brake)	0.832
Max (Brake)	2.554

Control Buttons: Start, Reset, **Setup**, Stop.

Click "Stop" to complete the ADC input setup as screenshot 7-36.

Screen 7-36. ADC Input Setup Complete

The screenshot displays the 'ADC Input Setup Complete' window. The interface is identical to Screenshot 7-35, but the 'Stop' button is now highlighted with a red box, indicating the setup is complete.

Parameter	Value
Min Throttle Voltage(V)	0.100
Mid Throttle Voltage(V)	1.650
Max Throttle Voltage(V)	3.200
Min Brake Voltage(V)	0.100
Max Brake Voltage(V)	3.200
Cruising Control Mode	None
Time for Enter Cruising(s)	10.0
Realtime Throttle ADC(V)	0.864
Realtime Brake ADC(V)	0.834
Min (Throttle)	0.860
Mid (Throttle)	1.706
Max (Throttle)	2.552
Min (Brake)	0.832
Max (Brake)	2.554

Control Buttons: Start, Reset, Setup, **Stop**.

7.8.4.3 ADC Control Mode Setup

Select the appropriate ADC Control Mode per your needs as screenshot 7-37.

Screenshot 7-37. ADC Control Mode

ADC

Control Mode
None

- None
- Current
- Current Bidirectional
- Current LMH-Gear
- Current Brake
- Current Brake LMH-Gear
- Current R-Gear
- Current R-Gear LMH-Gear
- Current R-Gear Brake
- Current R-Gear Brake LMH-Gear

Max Throttle Voltage(V)
3.200

Min Brake Voltage(V)
0.100

Time for Enter Cruising(s)
10.0

Realtime Throttle ADC(V)
0.864

Min
0.860

Mid
1.706

Max
2.552

Realtime Brake ADC(V)
0.834

Min
0.832

Max
2.554

Start
Reset
Setup
Stop

- **None:** Does not respond to any ADC input signal.
- **Current:** Current control mode, ADC throttle only without brake.
- **Current Gear:** Current control mode, only enable ADC throttle and LM\H speed gear.
- **Current Brake:** Current control mode, enable ADC throttle and brake.
- **Current Brake Gear:** Current control mode, enable ADC throttle, brake and LM\H speed gear.
- **Current Reverse:** Current control mode, enable ADC throttle and reverse gear.
- **Current Reverse Gear:** Current control mode, enable ADC throttle, reverse gear and LM\H speed gear.
- **Current Reverse Brake:** Current control mode, enable ADC throttle, brake and reverse gear.
- **Current Reverse Brake Gear:** Current control mode, enable ADC throttle, brake, reverse gear and LM\H speed gear.
- **Duty:** Duty cycle mode, ADC throttle only without brake
- **Duty Bidirectional:** bidirectional duty cycle control mode, ADC throttle only without brake
- **Duty R-Gear:** reverse duty cycle control mode, ADC throttle only without brake

7.8.4.4 Cruising Control Mode Setup

Cruising Control Mode: In ADC control mode, add Cruising Control Mode setup in different situations as shown in 7-38.

Screenshot 7-38.Cruising Control Mode

The screenshot displays the 'ADC' control mode configuration window. At the top, 'Control Mode' is set to 'None'. Below this, several voltage and time parameters are adjustable: Min Throttle Voltage (0.100V), Mid Throttle Voltage (1.650V), Max Throttle Voltage (3.200V), Min Brake Voltage (0.100V), Max Brake Voltage (3.200V), and Time for Enter Cruising (10.0s). The 'Cruising Control Mode' dropdown menu is open, showing four options: 'None', 'Self-Locking Button', 'Resetting Button', and 'Keep Throttle ADC for a While' (with a value of 0.004s). A red rectangle highlights this dropdown menu. A 'Start' button is located at the bottom right of the configuration area.

Parameters description

■ **None:** OFF the Cruising Control Mode

■ Self-Locking Button

Self-Locking Button: Connect the PPM and GND pins for the self-locking button. Press the switch button to enter into cruise mode, and then re-press the button to bump out to end the cruise mode.

Note: The cruise mode can be entered only in status of throttle signal detected or motor running with RPM; The cruise mode will stop if receive a brake signal in ADC control mode.

■ Resetting Button

Resetting Button: Connect the PPM and GND pins for the self-locking button. Press once to enter into the cruise mode, and press again to end the cruise mode.

Note: The cruise mode can be entered only in status of throttle signal detected or motor running with RPM; The cruise mode will stop if receive a brake signal in ADC control mode.

■ Keep Throttle ADC for a While

Keep Throttle ADC for a While: Enter the cruise mode by holding the throttle for a period of time. Keep the throttle at desired level for a period of time and then enter into the cruise mode. Stop the cruise mode after detecting any throttle change or brake signal. The throttle hold time can be set per users' needs via Time for Enter Cruising in ESC tool.

Note: The cruise mode will stop if receive a brake signal in ADC control mode.

◆ Time for Enter Cruising

If the cruise control mode is set to Keep Throttle ADC for a While, you can set the Time for Enter Cruising per your needs. Adjustable range: 3s to 30s, and the default is 10s as shown in 7-39.

Screenshot 7-39. Time for Enter Cruising Setup

The screenshot displays the 'ADC' control mode configuration window. It includes several adjustable parameters:

- Control Mode:** Set to 'None'.
- Min Throttle Voltage(V):** 0.100
- Mid Throttle Voltage(V):** 1.650
- Max Throttle Voltage(V):** 3.200
- Min Brake Voltage(V):** 0.100
- Max Brake Voltage(V):** 3.200
- Cruising Control Mode:** Set to 'None'.
- Time for Enter Cruising(s):** 10.0 (highlighted with a red box).

7.8.5 PAS

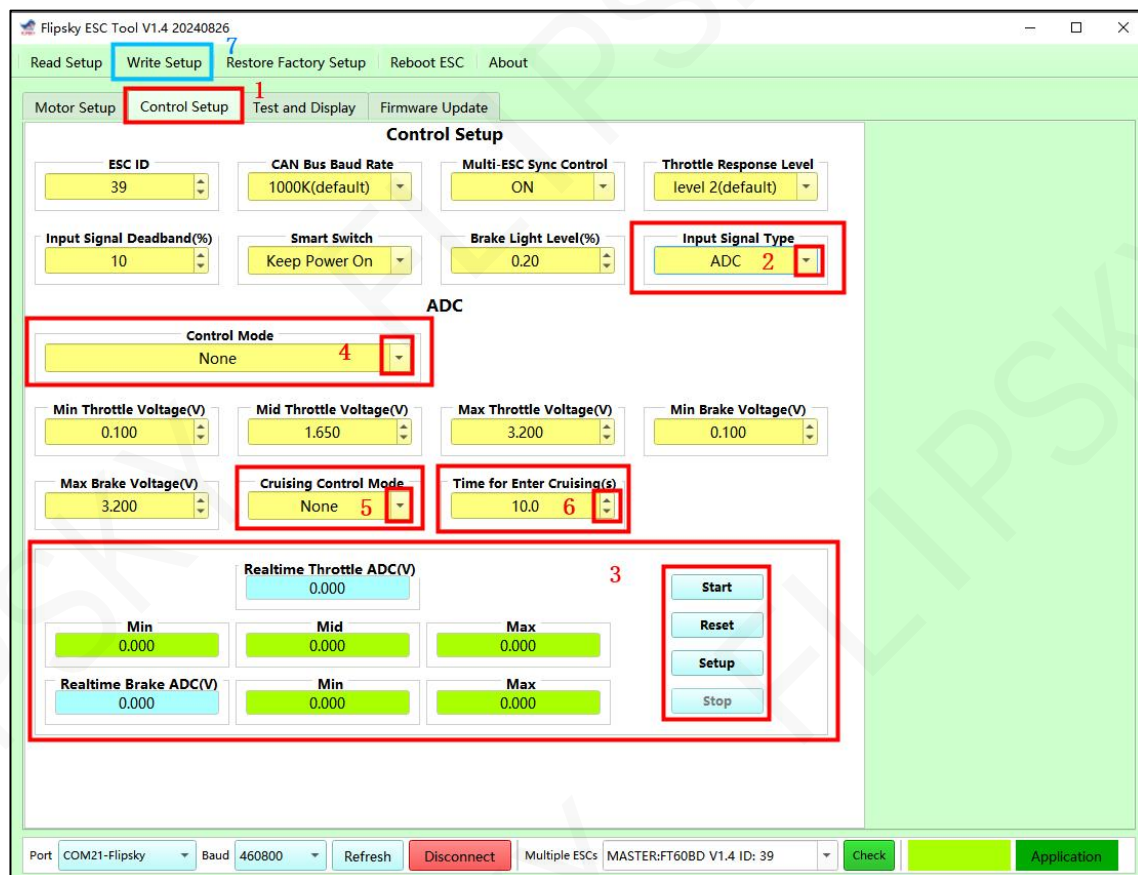
7.8.5.1 PAS Description and Setup

PAS Description:

- (1) It's compatible with most of PAS cadence sensors in the market, which mainly include below three types (refer to 7.8.5.3).
- (2) Enable FT ESCs to be compatible with Flipsky TFT screen display which supports real-time data like speed, voltage etc. read and PAS speed gear levels setup.
- (3) Support Brake Signal function: The ESC will cut off power output immediately once receiving the brake signal; When the control mode set to enable Brake Signal, pin GEAR_R and pin GEAR_L are used to detect the brake signal and are active at Vol. That is, when the pin GEAR_R or the pin GEAR_L detects a Vol(GND), the ESC will cut off the power output immediately, and disable the throttle gear.

Setting steps as below screenshot 7-40:

Screenshot 7-40.PAS Input Signal Type

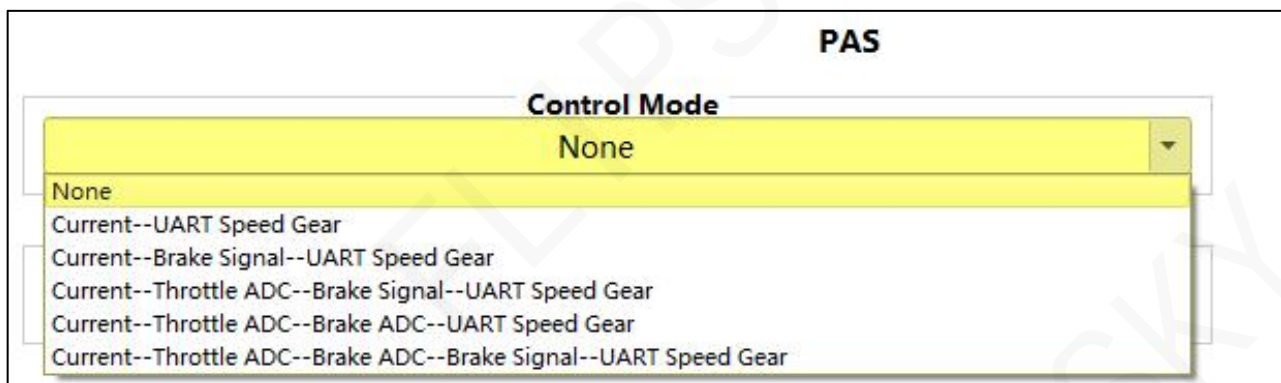


- Step 1: Click "Control Setup" to enter the setup interface.
- Step 2: Switch the "Input Signal Type" to "PAS"
- Step 3: Need to do ADC throttle input setup for the first time setup of PAS input signal (see details in 7.8.4.2)
- Step 4: Select the appropriate PAS Control Mode per your needs (see details in 7.8.5.2)
- Step 5: Select the appropriate PAS Sensor Type per your needs (see details in 7.8.5.3)
- Step 6: Set the PAS parameters (see details in 7.8.5.4)
- Step 7: Click the upper left menu "Write Setup" to save the setup.

7.8.5.2 PAS Control Mode Setup

Select the appropriate PAS Control Mode per your needs as screenshot 7-41:

Screenshot 7-41.PAS Control Mode



- None: No respond to any PAS input signal
- Current--UART Speed Gear: Current control mode: UART Speed Gear--Use ESC UART port to connect the Flipsky TFT screen display to switch different speed gear levels
- Current--Brake Signal--UART Speed Gear: Current control mode: Brake Signal--The ESC will cut off power output immediately once receiving the brake signal; UART Speed Gear--Use ESC UART port to connect the Flipsky TFT screen display to switch different speed gear levels.
- Current--Throttle ADC--Brake Signal--UART Speed Gear: Current control mode: enable throttle ADC control; Brake Signal--The ESC will cut off power output immediately once receiving the brake signal; UART Speed Gear--Use ESC UART port to connect the Flipsky TFT screen

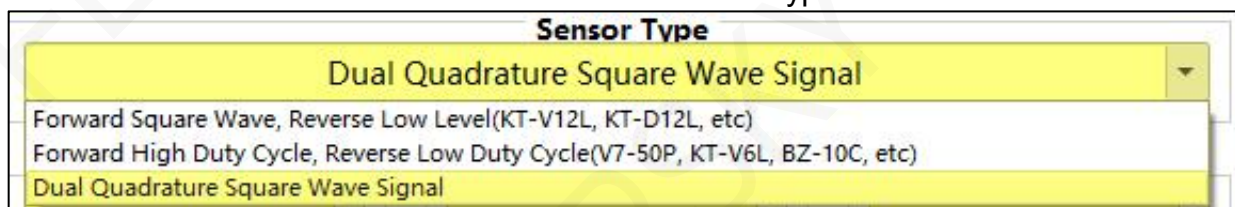
display to switch different speed gear levels.

- Current--Throttle ADC--Brake ADC--UART Speed Gear: Current control mode: enable throttle ADC and Brake ADC control; UART Speed Gear--Use ESC UART port to connect the Flipsky TFT screen display to switch different speed gear levels.
- Current--Throttle ADC--Brake ADC--Brake Signal--UART Speed Gear: Current control mode: enable throttle ADC and Brake ADC control; Brake Signal--The ESC will cut off power output immediately once receiving the brake signal; UART Speed Gear--Use ESC UART port to connect the Flipsky TFT screen display to switch different speed gear levels.

7.8.5.3 PAS Sensor Type

Select the appropriate PAS Sensor Type per your needs as screenshot 7-42.

Screenshot 7-42.Sensor Type



- Square wave output when rotates in forward direction; VOL output when rotates in reverse.
- High duty cycle rectangular wave output when rotates in forward direction; Low duty cycle rectangular wave output when rotates in reverse.
- Dual quadrature square wave signal output.

7.8.5.4 PAS Parameters Setup

PAS parameters: PAS Throttle, Min Pedal RPM, Max Pedal RPM and Sensor Magnets as screenshot 7-43.

Screenshot 7-43.PAS Parameters Setup



- PAS Current Throttle: PAS max current output percentage (adjustable range: 1-95%)

The maximum PAS current output will be limited to this setting percentage of the motor current.

- Min Pedal RPM: the minimum PAS pedal RPM (adjustable range: 1-200RPM)

The ESC starts current output when the pedal RPM is over the mini setting value. Otherwise, the current output is zero, no current output.

- Max Pedal RPM: the maximum PAS pedal RPM (adjustable range: 1-500RPM)

The ESC current output stops increasing if the pedal RPM is over the max setting value.

- Sensor Magnets: sensor magnets q'ty (adjustable range: 1-32)

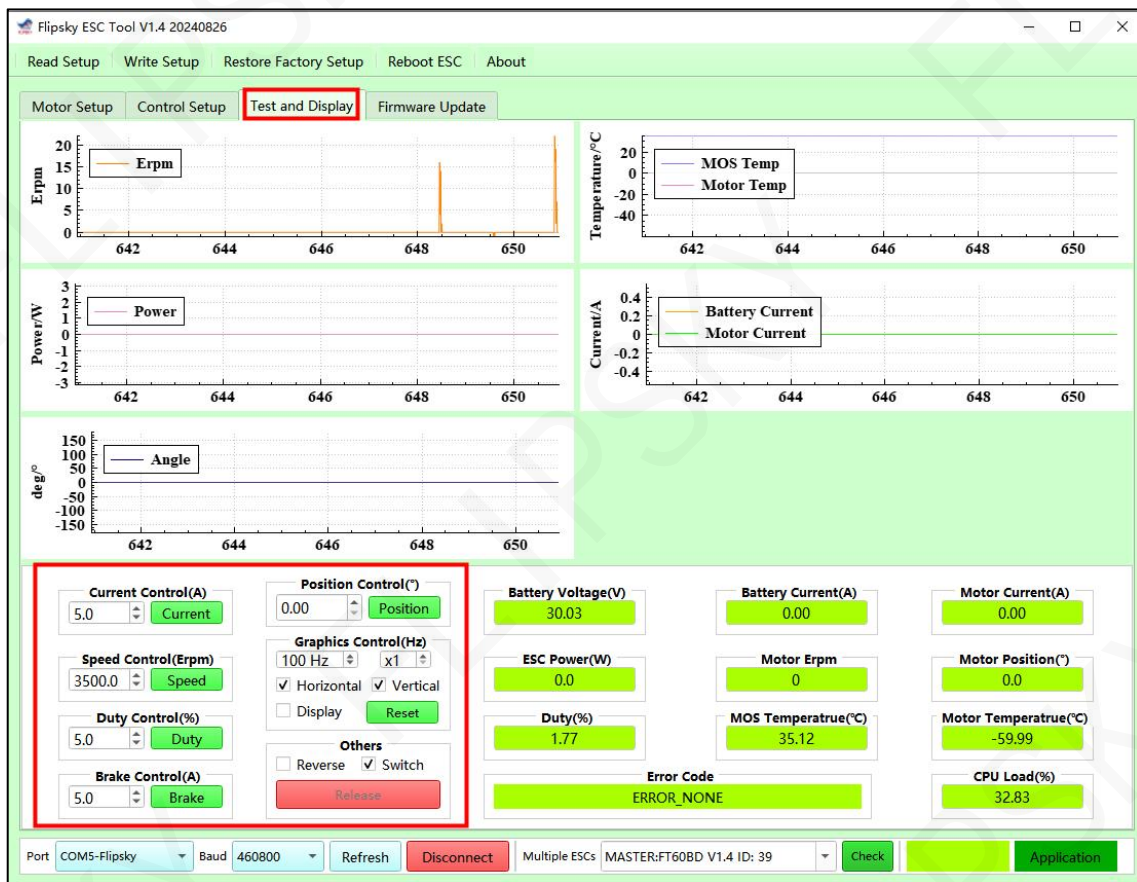
PAS magnets q'ty, i.e.: 12pcs magnets have 12 pulses each turn circle.

8 Test and Display

8.1 Flipsky ESC Tool Button Control

Switch the "Input Signal Type" to "OFF" if you want to control the ESC through the Current button, Duty button or Brake button in "Test and Display" interface as below screenshot 8-1.

Screenshot 8-1. Control Buttons in Test and Display Interface



8.1.1 Button Control Setup

How to use the buttons to control the ESC?

- Drive the motor: Click the "Current"-current control mode (torque control mode), "Speed"-speed control mode, "Duty"-duty cycle control mode (voltage control mode) and "Position"-position control mode buttons to drive. During the driving process, the current, speed, duty cycle and position angle can be adjusted according to needs as shown in 8-2:

Screenshot 8-2 Four Control Modes Setup

The screenshot displays the 'Four Control Modes Setup' interface. It features four primary control modes, each with a numeric input field and a corresponding green button: 'Current Control(A)' (5.0, Current), 'Speed Control(Erpm)' (3500.0, Speed), 'Duty Control(%)' (5.0, Duty), and 'Brake Control(A)' (5.0, Brake). A fifth mode, 'Position Control(°)', is also present with a numeric input (0.00) and a green button (Position). Below these are the 'Graphics Control(Hz)' settings, including a frequency input (100 Hz), a multiplier (x1), checkboxes for 'Horizontal' and 'Vertical' (both checked), a 'Display' checkbox (unchecked), and a 'Reset' button. The 'Others' section at the bottom includes a 'Reverse' checkbox (unchecked), a 'Switch' checkbox (checked), and a large red 'Release' button. The 'Current', 'Speed', 'Duty', and 'Position' control sections are highlighted with red rectangular boxes.

Note: The Position Control is applicable in Encoder Control Mode only.

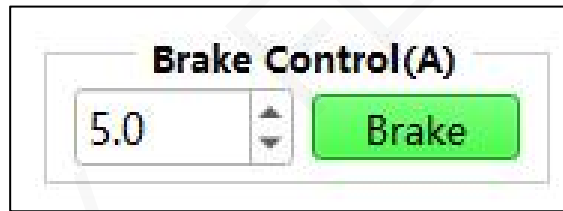
- Motor Rotation Direction Reverse: Tick "Reverse" to change the motor running direction as screenshot 8-3.

Screenshot 8-3. Reverse Direction

The screenshot shows the 'Others' section of the interface. It contains two checkboxes: 'Reverse' (unchecked) and 'Switch' (checked). Below these is a large red 'Release' button. The 'Reverse' checkbox is highlighted with a red rectangular box.

- Brake: Click "Brake" to enter the brake setup as screenshot 8-4.

Screenshot 8-4. Brake

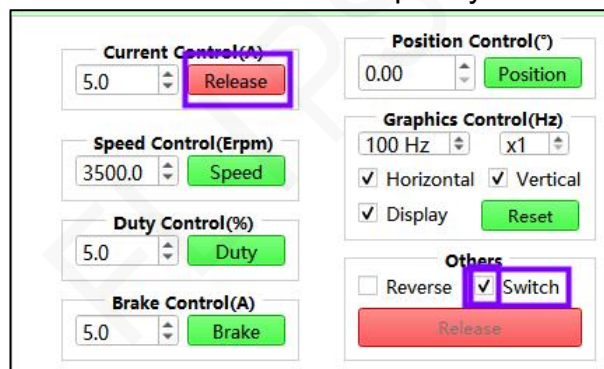


Notes: If using the DC power supply, please do not brake in high speed. Otherwise, the large back EMF will damage the ESC or even the power supply when the motor brakes in a high speed mode.

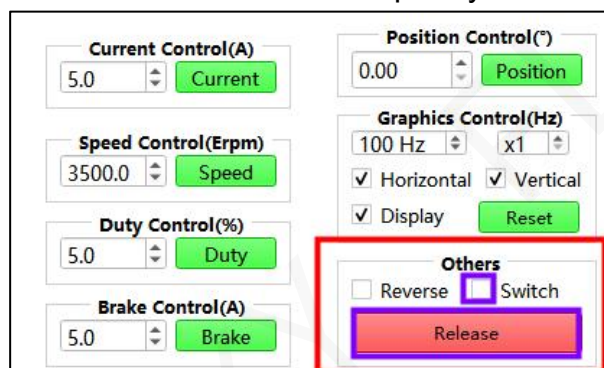
- Stop: Click "Release" to stop the motor running. Two ways available to stop the motor running: Tick "Switch" and click the "Current" under Current Control, then the "Current" will switch to "Release". Click the "Release" to stop the motor rotating as screenshot 8-5.

If not tick the "Switch" and it's "Current" under Current Control, the motor can only be stopped through the "Release" under the "Others" as screenshot 8-6.

Screenshot 8-5.Stop Way 1



Screenshot 8-6.Stop Way 2



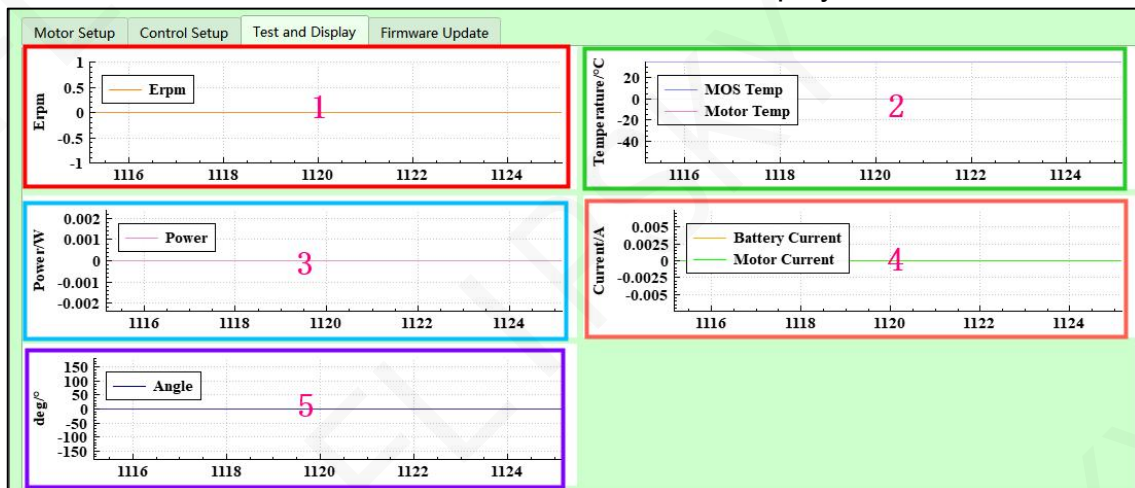
8.2 Data Analysis

8.2.1 Data Curve Display and Setup

There are 5 real-time data graphs as below screenshot 8-7:

- Erpm (Erpm)
- ESC MOSFET and motor temperature curve (Temperature/°C)
- Power curve (Power/W)
- Battery and motor current curve (Current/A)
- Motor deflection angle curve (deg/°)

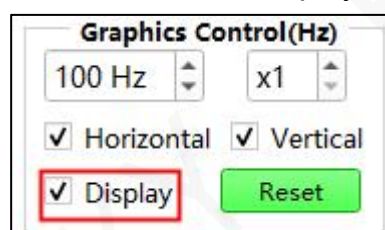
Screenshot 8-7. Real-time data Curve Display Interface



Data Curve Setup Steps: Realtime Data Display

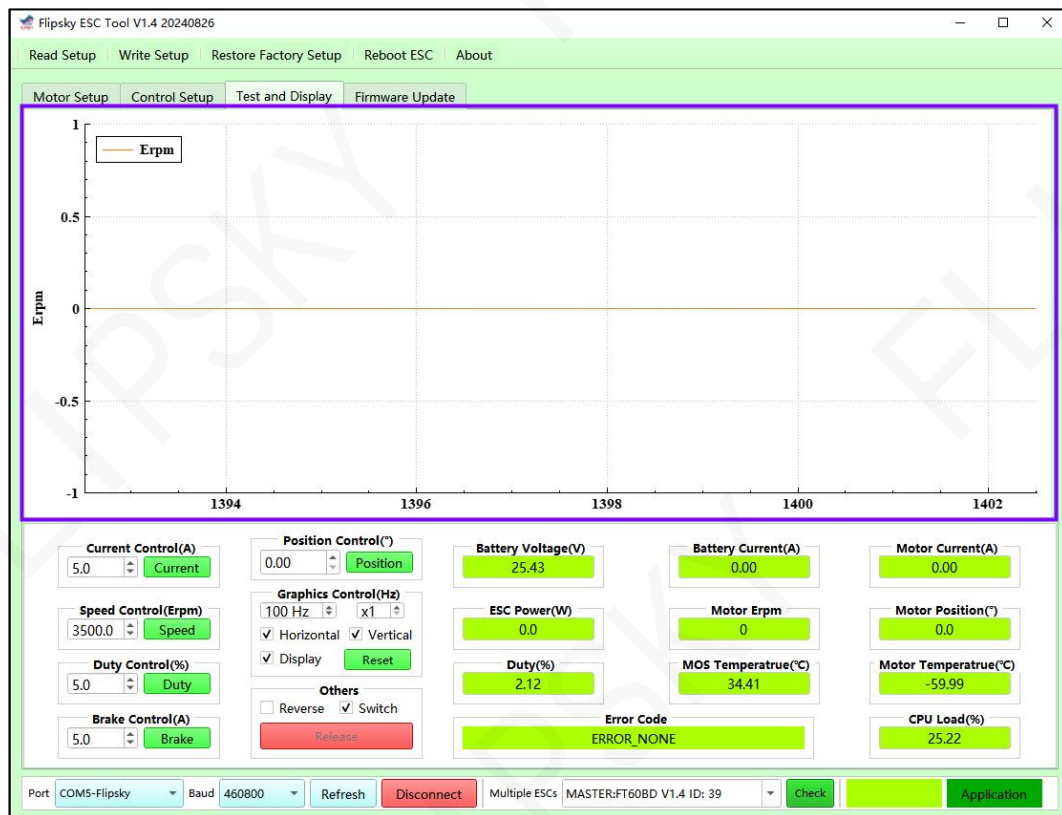
- Activate the graph: Tick "Display" to activate graphics data display as screenshot 8-8.

Screenshot 8-8. Data Display Setup



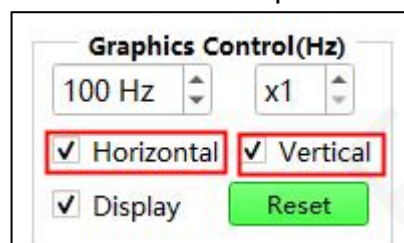
- Graph enlargement: move the mouse cursor in the graph and left-click twice to enlarge it, and restore it in the same way as below screenshot 8-9.

Screenshot 8-9. Graph Enlargement



- Increase or decrease the graph horizontal and vertical axis values: Tick both "Horizontal" and "Vertical" (as screenshot 8-10), move the mouse cursor into the graph and then scroll the mouse wheel to increase or decrease.

Screenshot 8-10. Graphics Control



Notes:

Tick the "Horizontal" and not tick the "Vertical" if you want to increase or decrease the values of horizontal axis only;

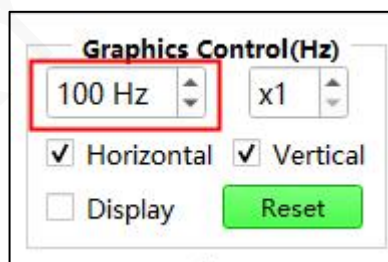
Tick the "Vertical" and not tick the "Horizontal" if you want to increase or decrease the values of

vertical axis only;

Tick both the "Horizontal" and the "Vertical" if you want to change values of both axis.

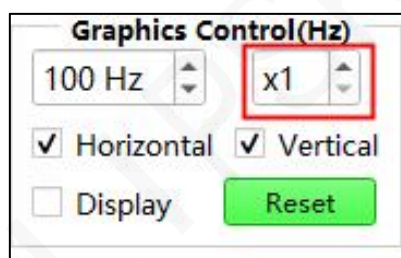
- The sampling frequency setup: default 100Hz, which means that the data samplings are 100 times per second. Adjustable setup range: 50Hz~1000HZ as screenshot 8-11.

Screenshot 8-11. The sampling frequency setup of graphics



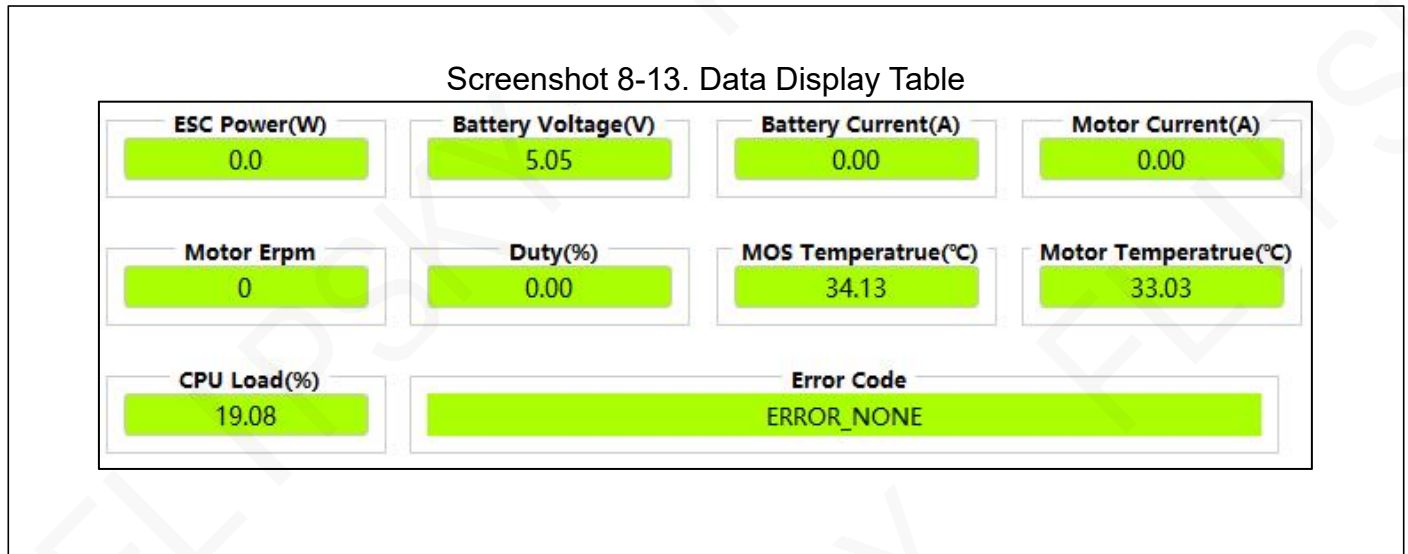
- Curve play speed setup: default "X1", adjustable range 1~10 times as screenshot 8-12.

Screenshot 8-12. Curve play speed adjustment



8.2.2 Realtime Data

Data display tables are listed under the graphics curve for better data analysis as screenshot 8-13.

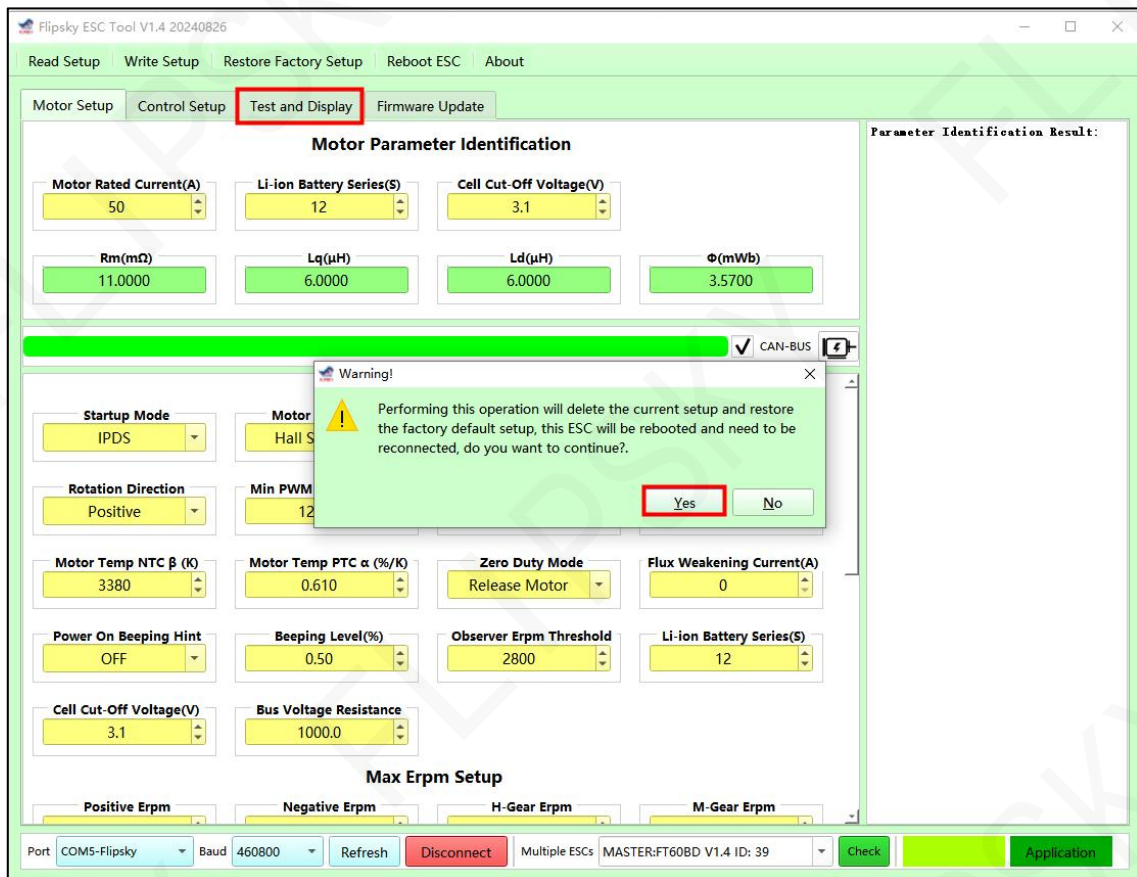


- ESC Power(W): ESC power display
- Battery Voltage(V): battery voltage display
- Battery Current(A): battery current display
- Motor Current(A): motor current display
- Motor Erpm: motor electric rpm display
- Motor Position: The motor deflection angle position display
- Duty(%): Duty cycle percentage display
- MOS Temperature (°C): ESC mosfets temperature display
- Motor Temperature (°C): motor working temperature display
- CPU Load(%): CPU loading percentage display;
- Error Code: fault code display

9 Restore Factory Default Setup

If meeting with some problems like parameters setup failure, ESCs connection failure, motor parameter identification failure etc., please try to restore the factory default setup as screenshot 9-1.

Screenshot 9-1. Restore Factory Default Setup



10 Version

Table 1. File Version

Date	Version	Contents Change
2023.09.06	V1.0	
2023.11.08	V1.1	1. Fixed the problem that the Hall parameter cannot pass when the load is too heavy. 2. Optimize the parameter adjustment procedure. All interfaces are shielded during parameter adjustment to prevent other signals from affecting parameter adjustment. 3. Optimize the dead zone setting mode in ADC mode, and fix the problem that some gamepads may not start smoothly when used; 4. Rectify the fault that the electrical connection to the upper computer fails.
2024.01.19	V1.2	1. Add FT ESC controller: FT85BD 2. Add Smart Switch function supporting slide poweron and auto turn off. 3. Add Extra Interface setup to enable front light, rear brake light and horn etc. for ESC FT85BD under UART control mode. 4. Add Slow-Release setup to enable FT series ESCs to maintain a certain current output after releasing the throttle under the UART input signal with smoother starting process. 5. Add Duty cycle, Duty Bidirectional and Duty R-Gear under ADC input signal. 6. Add Zero Duty Mode

		7. Optimize the motor flux linkage detection during the motor identification 8. Optimize the throttle gear speed level setup. 9. Optimize the realtime data curve display function.
2024/04/22	V1.3	1. Add FT ESC controller: FT85RS 2. Optimize the motor flux linkage detection for some small motors with large resistance. 3. Optimize real-time data acquisition protocol (Notes: the remote VX4 needs to update firmware V1.3.0, otherwise the real-time data will display abnormally and the parameter adjustment results will display abnormally.) 4. The minimum cut-off voltage of lithium battery can be set to 2.6V; Support LiFePO4 battery. 5. Add PAS Function: (1) It's compatible with most of PAS cadence sensors in the market, which mainly include below three types: 1) Forward Square Wave, Reverse Low Level like KT-V12L, KT-D12L etc. 2) Forward High Duty Cycle, Reverse Low Duty Cycle like V7-50P, KT-V6L, BZ-10C etc. 3) Dual quadrature square wave signal output: the A-phase signal leads the B-phase signal when rotating in the forward direction, while the A-phase signal lags behind the B-phase signal when rotating in the backward / reverse direction. (2) Enable FT ESCs to be compatible with Flipsky TFT screen display which supports real-time data like speed, voltage etc. read and PAS speed gear levels setup. (3) Support Brake Signal function: The ESC will cut off power output immediately once receiving the brake signal; When the control mode set to enable Brake Signal, pin

		GEAR_R and pin GEAR_L are used to detect the brake signal and are active at Vol. That is, when the pin GEAR_R or the pin GEAR_L detects a Vol(GND), the ESC will cut off the power output immediately, and disable the throttle gear. Optimize Flipsky ESC Tool Functions: 1. Add the parameters settings and control modes for waterproof esc controller FT85RS. 2. Add PAS Function
2024/8/24	V1.4	1. Add FT ESC controller: FT60BD 2. Optimize the motor voltage detection accuracy by adding motor "Bus Voltage Resistance" setup. 3. Add Flux Weakening Current setup to allow the motor to operate at speeds above its rated speed. 4. Add more Motor Sensor Port choices and enable encoder control choices setup like AS5047 (SPI) and ABI. 5. Add speed control PID mode. 6. Add position control PID mode (for encode control status only). 7. Add more choices for motor temperature sensor types such as NTC-10K-25°C, NTC-100K-25°C, PT1000, PTC-1K-100°C, KTY83 series, KTY84 series etc. 8. Add ESC power on beeping alert ON/OFF setup. Also, the beeping voice level percentage can be adjusted. 9. Add Brake Light Level setup to enable the light brightness to be adjusted during the braking process (available for FT60BD and FT85BD only at present). 10. In UART control mode, add Assisted Turnaround Mode setup like auxiliary turning function, fixed-point turning for 2WD mode and tank turning for 4WD mode. 11. In UART control mode, add Brake-Parking Mode ON/OFF setup to support parking mode by pulling remote controller throttle gear/bar (achieved by transferring current into the motor D axis) 12. In UART control mode, add Off-Road Climbing Mode ON/OFF setup to switch duty cycle control and allow users to pass off-road climbing process easily by

		controlling speed of the four wheels synchronously. 13. In ADC control mode, add Cruising Control Mode setup in three different situations. (1) Self-locking switch (2) Reset switch (3) Throttle hold Optimize Flipsky ESC Tool Functions: 1. Add the parameters settings and control modes for waterproof esc controller FT60BD. 2. Add speed control and position control modes in Test and Display interface; add position data display.